The Effects of Early-Childhood Home Visits on Grandparental Childcare and Employment

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Individuals who experienced maltreatment in childhood face an elevated risk of maltreating their own children, but supportive relationships can moderate this intergenerational transmission. Parents with such histories, however, are also the least likely to have a strong support system. A parent's own history cannot be undone, yet the support available to them may still be malleable. In this study, I examine the effect of Supportive Parenting, a Dutch early-childhood intervention that aims to prevent child maltreatment and consists of six nurse home visits in the first years after birth, on grandparental involvement in childcare, the most common source of informal care. I restrict my analyses to families that are eligible for Supportive Parenting due to the mother experiencing maltreatment during her childhood. With data from a randomised controlled trial that started in 2001 linked to administrative records, I document that mothers with a history of childhood maltreatment face poorer socio-economic outcomes and lower perceived social support than mothers without this history. I show that Supportive Parenting increases grandparental childcare among these mothers, driven by paternal grandparents, among whom it prevents the decline in childcare observed in the control group. I also find evidence that Supportive Parenting increases paternal employment, with no evidence of effects on maternal or grandparental employment. Effects on childcare are concentrated among families with stronger baseline support, suggesting that the visits help parents activate latent support rather than build new relationships. I find no evidence that the effect operates through agreement between parents and grandparents, partner contributions, confiding about parenting difficulties, or earlier grandparental retirement. My findings do not rule out that the visits work by helping parents identify whom to approach for support and how to do so.

Keywords: nurse home visit, randomised controlled trial, maltreatment, grandparental childcare, employment

JEL codes: J12, J13, J22

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The non-public administrative microdata used in this paper are available via remote access to the Microdata services of Statistics Netherlands (CBS). As stipulated in the data agreement, Statistics Netherlands previewed the findings of this study prior to publication to ensure that privacy sensitive, individual-specific information is not revealed. The data from this study can only be applied for through a government data sharing portal of Statistics Netherlands (https://www.cbs.nl/en-gb/our-services/customised-services-microdata/microdata-conducting-your-own-research). Under certain conditions, these microdata are accessible for statistical and scientific research. For further information: microdata@cbs.nl.

1 Introduction

Individuals who experienced maltreatment in childhood face an elevated risk of maltreating their own children (Assink et al., 2018). This risk is not fixed: safe, stable, nurturing relationships can moderate the intergenerational continuity of child maltreatment (Schofield et al., 2013). Yet the parents for whom such relationships are most protective are also the least likely to have them. Adverse childhood experiences are associated with strained relationships between adult children and their parents (Kong and Martire, 2019), and, as I show, mothers who experienced maltreatment in their own childhood are less likely to feel accepted by their family and less likely to feel sufficiently supported than other mothers. A parent's own history of maltreatment cannot be undone, but the support available to them may still be malleable. Grandparents are a particularly relevant source of such support, as they are a common provider of informal childcare. Across European countries, between 24 and 60 percent of grandparents provide childcare (Zanasi et al., 2023).[1] Home visiting programmes often aim to strengthen parents' support networks, yet evaluations of these programmes have focused on outcomes for children and mothers (Heckman et al., 2017; Doyle, 2020; Sandner, 2019). Little is known about whether they change the behaviour of the extended family. In this paper, I study Supportive Parenting (SP), a Dutch programme of six nurse home visits in the first two years after birth, in families where the mother experienced maltreatment in childhood. I use data from a randomised controlled trial of children born in 2001 and 2002, linked to administrative records of their parents and grandparents. I find that SP prevents a decline in childcare provided by paternal grandparents, such that 24 months after birth, 32 percent of treated families receive childcare from paternal grandparents, compared to 18 percent of control families. I find no evidence of effects on childcare from maternal grandparents. SP also increases fathers' employment by 1.5 months over the first three years after birth.

Supportive Parenting is a Dutch early-childhood intervention that consists of six home visits by trained nurses in the first 18 months after birth. The intervention aims to prevent parenting problems and child maltreatment. Eligibility is determined by a universal screener to identify parents at risk of parenting problems. Parents with a newborn receive the screener, along with information on SP. Two weeks after birth, each family with a newborn in the Netherlands receives a home visit by a preventive youth healthcare nurse. In the catchment area, the nurse collects the eligibility screener during this visit. As part of the screener, parents answer four questions about childhood maltreatment, where self-reported maltreatment in childhood by one parent immediately qualifies a family for the intervention. For these parents, the intervention aims to mitigate the intergenerational transmission of child maltreatment[2] by teaching parents about child development, reflecting with them on their childhood experiences, and helping them build a strong support system[3].

[1] In the Netherlands, 60 percent of grandparents are involved in some childcare and 35 percent provide weekly childcare for a grandchild (Zanasi et al., 2023).

[2] A substantial literature finds an intergenerational association in child maltreatment (Assink et al., 2018; Madigan et al., 2019).

[3] Safe, stable, nurturing relationships can moderate the intergenerational continuity of child maltreatment

SP may affect grandparental childcare by reducing the costs of coordinating an informal transfer between parents and grandparents. In a cooperative bargaining framework, childcare can be viewed as a potentially mutually beneficial intergenerational transfer whose realisation depends on the costs of reaching and sustaining such agreements. Compton and Pollak (2026) emphasise the role of transaction costs in determining whether mutually beneficial exchanges within extended families are realised. Similarly, Roeters et al. (2020) show how coordination and trust can shape informal care arrangements. This mechanism is relevant to SP because activating parents' social networks is an explicit component of the intervention (Horrevorts et al., 2015). By helping parents identify and mobilise available sources of support, SP may reduce information and coordination frictions involved in arranging informal childcare. The intervention does not need to change grandparents' preferences or availability of time; it may instead make an existing source of childcare easier to mobilise. The effect may be particularly pronounced for paternal grandparents given the population targeted by SP. The intervention is offered to mothers with a history of childhood maltreatment, and for some mothers in the sample, their own parents may have been involved in this maltreatment or have failed to protect them. Paternal grandparents may therefore represent an alternative source of childcare.

In addition to analysing the effects of SP on grandparental childcare, I explore the effects on paternal and grandparental employment. Consistent with time-allocation theory, individuals face a fixed amount of available time that must be allocated across market work and non-market activities, including childcare, implying that time devoted to one activity carries an opportunity cost in terms of other uses of time (Becker, 1965). An increase in available childcare from grandparents relaxes parents' time constraint, freeing time previously spent on childcare for reallocation toward market work. Consistent with this mechanism, prior research has documented a positive association between grandparental childcare and maternal labour market participation (Geurts et al., 2015). Bratti et al. (2018) and Posadas and Vidal-Fernandez (2013) show causal evidence that grandparents' availability for childcare increases maternal labour supply. Although these papers do not analyse paternal labour market outcomes, research on the effects of formal childcare access on paternal employment shows limited or no effects (Andresen and Havnes, 2019; Brewer et al., 2022).

The same time-allocation logic implies a potential trade-off for grandparents. Grandparents' time is similarly constrained, where providing additional childcare requires reallocating time away from other activities, including paid work. Existing literature warns that an increase in parental labour market participation might therefore come at the cost of grandparental employment, suggesting a trade-off between grandparental childcare provision and grandparental labour market participation. Prior research consistently shows negative labour market effects of becoming a grandmother, including reductions in earnings and hours worked, as well as higher retirement probabilities (Andersen et al., 2025; Backhaus and Barslund, 2021; Gørtz et al., 2025; Joo et al., 2024; Malisa, 2024; Frimmel et al., 2022; Rupert and Zanella, 2018; Karademir et al., 2023). In contrast, the evidence for grandfathers shows little or no effects.

(Schofield et al., 2013).

Andersen et al. (2025) and Gørtz et al. (2025) find that grandparenthood reduces grandfathers' earnings, but to a lesser extent than grandmothers' earnings, whereas Joo et al. (2024) find no evidence of effects on grandfathers' earnings. Backhaus and Barslund (2021) find no effect on grandparents' labour supply.

To analyse the effects of SP, I use data from a randomised controlled trial (RCT) involving children born between October 2001 and October 2002. I use survey data collected during the South Holland Supportive Parenting experiment and administrative data. The participating children are linked to administrative records in a secured environment, where I connect them to their parents and grandparents. Grandparental childcare is measured in surveys completed when the child is 6 weeks, 12 months, and 24 months old. Because the 6-week survey is completed before the first home visit, it serves as a pre-treatment baseline. Employment data comes from tax registers available in the administrative data.

Of the 500 families that participate in the RCT, I focus on a subset of families that were eligible due to maternal childhood maltreatment (N=217), of which 111 were randomly assigned to the control group and 106 to the treatment group. Compliance is high: 94 percent of families in the treatment group received four or more of the six home visits, and no families in the control group received home visits. Baseline characteristics are balanced between the two groups, with the exception of maternal age, as mothers in the treatment group are on average 1.3 years younger. I estimate intention-to-treat effects, controlling for baseline characteristics and using entropy weights to correct for potential non-random attrition in the survey outcomes.

In the paper, I first descriptively show that mothers who experienced childhood maltreatment have less favourable socio-economic outcomes and perceive lower social support than mothers without a maltreatment history. These mothers are less likely to be employed, more likely to be on benefits, less likely to have a college degree, and less likely to cohabit with the child's father. They are also less likely to report feeling accepted by their family and to be content with the level of general support.

Second, I find that, among families eligible due to maternal maltreatment, Supportive Parenting increases paternal grandparental childcare. At baseline, 32 percent of control families and 34 percent of treatment families report childcare from paternal grandparents. In the control group, this share declines to 20 percent at 12 months and 18 percent at 24 months. The estimated effects of SP in three specifications are 6.85-8.64 percentage points at 12 months and 11.87-15.22 percentage points at 24 months. SP prevents a decline in paternal grandparental childcare relative to baseline. I find no evidence of effects on maternal grandparental childcare.

Third, SP increases the number of months fathers are employed during the three years following birth, with no evidence of effects on mothers and paternal grandparents. For fathers, SP increases employment by 1.5 months over the first three years after birth, compared to a control group mean of 32.8 months. I find no evidence that SP affects the total number of months in which mothers or grandparents receive income from employment in the three years after the birth of their (grand)child.

To examine whether effects depend on families' baseline support and capacity to mobilise

support, I examine heterogeneity by four measures of baseline support (general support, family acceptance, partner involvement, and ability to ask for help) and by a composite index combining them. An increase in childcare can either be through an activation of latent baseline support, or by establishing new connections. If SP nurses help parents mobilise existing sources of informal support, its effects may be stronger where supportive relationships are already available and parents are able to draw on them. If SP nurses help parents make new connections, its effects may be stronger where baseline support is low. Consistent with the first interpretation, I find that the increase in paternal grandparental childcare is concentrated among families with higher baseline support. This pattern is consistent with nurses helping parents identify whom to approach for support and plan how to do so. For maternal grandparental childcare, there is some evidence of an effect among mothers who feel accepted by their family, the group for whom the mother's own parents are most likely to be a safe source of support, but no evidence of effects across the other measures of baseline support.

In addition, I examine four pathways through which SP could affect grandparental childcare: agreement between parents and grandparents on parenting, partner contributions to childcare and household tasks, confiding about parenting difficulties, and grandparents' pension receipt. First, SP may affect whether parents and grandparents agree about how the child should be raised. By providing information about child development and encouraging parents to reflect on their own parenting experiences, the intervention may help parents develop and communicate a clearer approach to parenting. Greater parent–grandparent agreement could reduce conflict over caregiving practices and facilitate cooperation in providing childcare (Bai et al., 2023). Second, SP may affect the allocation of childcare and household tasks between parents. I find that SP increases fathers' employment, which may reduce the time they have available for childcare at home. If paternal and grandparental childcare are substitutes, a reduction in fathers' childcare contribution could increase the demand for grandparental childcare (Meil and Rogero-García, 2015). Third, SP may encourage parents to confide more in others about parenting problems. Nurses explicitly help parents identify sources of support and discuss how they can ask others for help. Parents may consequently become more likely to turn to others when experiencing difficulties with their child. Being able to confide in one's partner, friends or family about difficult parenting situations can indicate a willingness to seek support when it is needed. Such support-seeking may facilitate the subsequent mobilisation of practical assistance, including grandparental childcare. Fourth, SP may affect grandparents' availability to provide childcare. If the intervention induces grandparents to enter retirement earlier, their opportunity cost of providing childcare may decrease. Pension reforms show that grandparental retirement can affect childcare availability and provision (Bratti et al., 2018; Kaufmann et al., 2022).

I find no evidence that the increase in grandparental childcare operates through these channels. However, at baseline, 95 percent or more of parents in the control group report that their partner helps with childcare and household tasks, and that they have someone to confide in about parenting difficulties. This leaves little scope for SP to raise these further. A difficulty with pension status is that it can also be obtained by reaching the statutory pension age. These

findings do not rule out a reduction in the information and coordination costs of mobilising existing support. In addition, these four channels do not capture whether parents become better able to identify whom to approach, ask for help, or coordinate an informal childcare arrangement with. This interpretation is consistent with the intervention's explicit focus on helping parents identify and mobilise sources of support, but I cannot test it directly with the available data.

This paper contributes to four literatures. First, it contributes to the literature on early-childhood home visiting by showing that the effects of a parenting intervention extend beyond treated parents and children to the extended family. Existing evaluations of home-visiting programmes primarily study outcomes for children and mothers (Heckman et al., 2017; Doyle, 2020; Sandner, 2019). Second, the paper contributes to the literature on informal childcare and family support by showing that an intervention targeted at families where the mother experienced childhood maltreatment can affect grandparental childcare. This is particularly notable because the effect is concentrated among paternal grandparents, despite the well-documented matrilineal advantage in grandparental childcare (Thomése and Liefbroer, 2013; Brunello and Yamamura, 2023). Third, the paper contributes to the literature on the intergenerational consequences of childhood maltreatment. A history of childhood maltreatment is associated with difficulties in relationships with parents and with intergenerational support in adulthood (Kong, 2016; Kong and Martire, 2019). While this literature documents disadvantages in intergenerational relationships, less is known about whether these relationships can respond to intervention. I show that support within these intergenerational relationships can change. Fourth, the paper speaks to the literature on the labour-market consequences of grandparental childcare. Prior work documents positive labour market effects on mothers (Bratti et al., 2018; Posadas and Vidal-Fernandez, 2013) and negative effects on grandmothers (Andersen et al., 2025; Backhaus and Barslund, 2021; Gørtz et al., 2025; Joo et al., 2024; Malisa, 2024; Frimmel et al., 2022; Rupert and Zanella, 2018; Karademir et al., 2023). My point estimates suggest a similar trade-off, but effects are not significantly different from zero. I find that SP increases fathers' employment by 1.5 months over the first three years after birth. This finding is consistent with greater access to informal childcare relaxing parents' time constraints.

The effects of Supportive Parenting have previously been studied by van der Voort et al. (unpublished), using the full RCT sample, which also includes families eligible on grounds other than maternal childhood maltreatment. They find effects on the household, including improved parenting attitudes, reductions in parents' hours worked, and increased birth intervals, as well as improved socio-emotional development among children. In contrast, this paper focuses on the subsample of 217 families whose eligibility was based on maternal childhood maltreatment and examines grandparental childcare, an outcome not studied by van der Voort et al. I also examine parental labour-market participation and find an increase in fathers' months employed. Van der Voort et al. find no detectable effect on fathers' labour-market participation in the first three years after birth in the full sample. This difference is consistent with extended-family support being particularly relevant for families with a history of maternal childhood

maltreatment.

2 The South Holland Supportive Parenting Experiment

2.1 Aim and development

The development of Supportive Parenting was initiated by medical professionals who observed many cases of possible child maltreatment, for example in the hospital. In a doctoral dissertation project, Bouwmeester (2006) and a team of experts developed Supportive Parenting, with the aim to prevent parenting problems and child maltreatment. They conducted a randomised controlled trial that started in 2001, the data from which I use in my analyses. Supportive Parenting has been further developed since the experiment and is currently offered to parents at risk of parenting problems in half of the Dutch municipalities.[4]

2.2 Eligibility

The Supportive Parenting experiment took place in the Northern part of South Holland in the Netherlands (see Figure A.1). All parents with a child born in the catchment area between October 1, 2001, and October 31, 2002, were asked to complete a screener questionnaire. This screener, the Instrument for early identification of Parents At Risk for child Abuse and Neglect (IPARAN), was sent to their home within two weeks after birth. The screener consists of 16 IPARAN items that might predict parenting problems. In the IPARAN, parents report on their support system, their expectations about parenthood, their mental health, their use of alcohol and drugs, and their childhood experiences. The IPARAN screener has good predictive validity (van der Put et al., 2017). All items and their scoring mechanism are further explained in Appendix B.

In this paper I focus on families eligible because the mother experienced maltreatment during her childhood. Childhood maltreatment is a predictor for parenting problems, as adults who experienced maltreatment during their childhood have higher odds of maltreating their own children than parents without such a history (Assink et al., 2018). To identify these parents, parents report on four IPARAN items about adverse childhood experiences related to child maltreatment:

1. I feel that my parents/carers hit me too much as a child.
2. As a child I generally felt safe in the vicinity of (one of) my parents/carers.
3. I was forcefully subjected to an unpleasant sexual experience before I was 16.
4. I feel that the relationship between my parents/carers was often violent when I was a child.

[4]In the Netherlands, municipalities decide which interventions to offer. Figure A.2 shows a map of the municipalities offering Supportive Parenting in 2026. An updated overview of all regions offering Supportive Parenting can be found here: `https://www.stevigouderschap.nl/voor-professionals/beheer`.

The items identify parents who experienced physical maltreatment (item 1), emotional maltreatment (item 2), sexual maltreatment (item 3), or were a witness to parental violence (item 4). When a parent reports maltreatment on at least one of these four items (answering yes to items 1, 3, or 4, or answering sometimes or never to item 2), their family is automatically eligible for the Supportive Parenting home visits. In my analysis sample, I include families who are eligible because the mother reported having experienced maltreatment in her childhood.

2.3 Content of the home visits

Supportive Parenting consists of home visits when children are 6 weeks, three months, six months, nine months, 12 months, and 18 months old. All six home visits are 90 minutes and are conducted by a trained nurse. During visits, nurses focus on child development, parents' experiences, and social support:

1. **Child Development.** The nurse explains the normal development of a child to help parents form realistic expectations of their child and identify problematic development. Nurses help parents to better understand the child's behaviour and give practical advice to respond appropriately.

2. **Experiences.** The nurse helps parents reflect on their childhood experiences. Adverse experiences, such as child maltreatment, might affect parenting and relational behaviours. The nurse helps parents mitigate negative responses to experiences by reflecting on their parenting behaviours in relation to their past experiences, recognising patterns, and improving parenting behaviours. Reflection increases relational competence and helps mothers differentiate between harmful and safe relationships.

3. **Social support.** The nurse helps parents build a strong support system. Together with parents, the nurse maps the current social network, explores potential sources of safe support, identifies barriers to asking for help, and develops concrete plans to strengthen the social support system. Where needed, the nurse connects parents with other parties, such as sports associations or professional support.

Within every domain, the nurse and parents decide where improvement is needed, set objectives, and write down steps to accomplish them. Nurses use a case file with questions to guide the process.[5] SP is tailored to each family and during the home visits nurses and parents reflect on progress and adjust plans accordingly.

[5]Examples of open questions are "How does it feel to be a parent? How do parents deal with these feelings?" and "Are the parents in contact with friends? Are these friends helping with childcare?".

3 Data and Methodology

3.1 Randomisation

All parents with a child born between October 1, 2001, and October 31, 2002, in the RCT catchment area were asked to complete the IPARAN screener. Of these families, 8,805 (99.0%) could be linked to administrative data. I refer to these families as all families in the catchment area.

Among all families in the catchment area, 4,846 families include at least one parent who completed the IPARAN items on childhood maltreatment. These families include 4,846 mothers and 4,762 registered fathers; of these, 4,834 mothers and 4,662 fathers completed the maltreatment items themselves. This sample is referred to as the screener sample.

Of the families in the catchment area, 1,441 were found to be eligible for Supportive Parenting. Of the eligible families in the catchment area, 500 wanted to participate in the RCT and were randomly assigned to the control group (N = 258) or the treatment group (N = 242). Randomisation was neither stratified nor clustered. This sample is referred to as the RCT sample.

In this paper, I restrict my analyses to a subsample of the RCT sample. This subsample contains the randomised families who are eligible for Supportive Parenting due to maternal childhood maltreatment (111 in control and 106 in treatment). I refer to this sample as the analysis sample.

3.2 Analysis sample

In this paper, I restrict my sample to families who are eligible for Supportive Parenting due to maternal maltreatment. Adverse childhood experiences are associated with strained intergenerational relationships (Kong and Martire, 2019), potentially limiting the involvement of grandparents in the child's life. Compared with families who are eligible for other reasons, such as mental health problems, mothers in these families report lower baseline levels of support and family acceptance, and these families receive less grandparental childcare at baseline (see Appendix D.3), leaving greater scope for improvement. Consequently, grandparental involvement is a more relevant and malleable aspect of the family environment in this group, making it a more prominent focus for Supportive Parenting nurses seeking to strengthen the family support system.

I focus on families eligible due to maternal, rather than paternal, childhood maltreatment, and I analyse maternal and paternal grandparental childcare separately, for two reasons. First, parents report whether they experienced maltreatment in their childhood, but not who the perpetrator was. It is therefore possible that the grandparents involved in childcare are responsible for the maltreatment that a parent experienced. By conditioning on maternal maltreatment and separating maternal from paternal grandparents, I can distinguish childcare provided by grandparents on the side of the parent who reported maltreatment from childcare provided by grandparents on the other side of the family. Because some fathers in the analysis sample also

report childhood maltreatment, I show in Figure E.6 that the results are similar when excluding these families. Second, I distinguish between maternal and paternal grandparental childcare because of the documented differences in childcare provision between maternal and paternal grandparents, with maternal grandparents being more involved than paternal grandparents (Danielsbacka et al., 2011; Eisenberg, 1988; Laham et al., 2005; Thomése and Liefbroer, 2013).

This leaves me with a sample of families eligible for Supportive Parenting due to maternal maltreatment (N = 217) and a sample of families eligible due to paternal maltreatment (N = 113). The sample of families eligible due to paternal maltreatment is too small for the analyses conducted in this paper, as shown by the power calculations in Appendix G. In addition, more data are available for mothers than for fathers. With the exception of the IPARAN screener, all surveys were completed by one parent, and the mother was the respondent in most cases (92–96% depending on the survey). This makes it difficult to analyse the channels through which Supportive Parenting operates in families eligible due to paternal maltreatment. Moreover, while all children in the RCT have a registered mother, not all have a registered father, resulting in complete administrative data for mothers but some missing data for fathers.

3.3 Data

I use survey data and administrative data to construct baseline variables and outcome variables. The data sources and variables are further described in this section.

Survey data

Both parents in the catchment area are asked to fill out the IPARAN screener within 14 days after birth. In addition, in the families enrolled in the RCT, one parent is asked to fill out a social support survey when the child is 6 weeks, 12 months, and 24 months old. The 6 week survey is filled out within 6 weeks after birth, after the eligible family is invited to participate in the experiment, but before the family is assignment to the treatment or control group and the first home visit. The respondent is most often the mother: in the analysis sample, the mother filled out the social support survey in 96, 95, and 92 percent of families at 6 weeks, 12 months, and 24 months, respectively. All questions are asked in Dutch. Table C.2 shows an overview of the surveys and their translated items used to construct the variables, as well as the time at which the data were collected. A brief overview of the variables is given here:

Sample identification

Parents maltreated during their childhood. Within 14 days after birth, both parents report whether they experienced physical, emotional, or sexual maltreatment, or witnessed parental violence, during their childhood. Families are included in the analysis sample if the mother reports having experienced maltreatment.

Descriptive and heterogeneity

General support. Within 14 days after birth, both parents report whether they would like more people around for support. I create a dummy for each parent equal to 1 if they report never wanting more support. The mother's support indicator is also used in heterogeneity analyses. For heterogeneity dummy variables, missing values are replaced by 0.

Family acceptance. Within 14 days after birth, both parents report whether they feel accepted by their family. I create a dummy for each parent equal to 1 if they report always feeling accepted. The mother's acceptance indicator is also used in heterogeneity analyses. For heterogeneity dummy variables, missing values are replaced by 0.

Partner involvement. Within 14 days after birth, the mother reports whether she expects her partner to be sufficiently involved in raising the child. I create a dummy variable equal to 1 if she indicates always expecting her partner to be sufficiently involved. For heterogeneity dummy variables, missing values are replaced by 0.

Ability to ask for help. Within 14 days after birth, the mother reports whether it is easy for her to ask for help or advice when she needs it. I create a dummy variable equal to 1 when she indicates that it is always or often easy for her to ask for help when needed. For heterogeneity dummy variables, missing values are replaced by 0.

Baseline and outcomes

Grandparental childcare. When the child is 6 weeks, 12 months, and 24 months old, one parent reports whether they receive help from their own parents and from their in-laws with raising the child, and whether they are the child's mother or father. From this, I construct dummies for maternal and paternal grandparental childcare. The 6-week measure serves as baseline; the 12- and 24-month measures are used as outcomes.[6]

Channels

Agreement on parenting. When the child is 6 weeks, 12 months, and 24 months old, one parent reports whether their own parents and their in-laws agree with their parenting approach. I construct dummies for maternal and paternal grandparental agreement.

Partner contribution. When the child is 6 weeks, 12 months, and 24 months old, one parent reports whether their partner helps with childcare and with household tasks. I construct a dummy for each.

Confiding about parenting difficulties. When the child is 6 weeks, 12 months, and 24 months old, one parent reports whether they can talk to their partner, and to a friend or family member, when their child is difficult. I construct a dummy for each.

Robustness

Support from others. When the child is 6 weeks, 12 months, and 24 months old, one parent reports whether they are in contact with other parents, neighbours, and friends. I construct a dummy for each.

[6]The Dutch term *opvoeding* refers to child-rearing broadly. The measure therefore captures grandparental help with raising the child, which includes but is not limited to looking after the child.

Administrative data

In addition to survey data, I use administrative data from Statistics Netherlands. With identifying information, such as birth dates and addresses, Statistics Netherlands linked the participants of the Supportive Parenting experiment to the administrative data in a secured environment. 99.0% of families were successfully linked.[7] Statistics Netherlands then encrypted the data to ensure the inability to identify the respondents.

With the administrative data, I link children to their registered parents and grandparents.[8] My sample is limited to individuals registered in the Personal Records Database (BRP), which includes persons who currently live in the Netherlands, have lived in the Netherlands, or have an administrative relationship with the Netherlands (for example due to cross-border employment). For individuals who were only in the BRP before October 1, 1994, the administrative data are incomplete. Table D.4 shows how many children, parents, and grandparents could be linked in the descriptive and analysis sample.

Descriptive

Age and birth month. With information on the birth year and month, I construct the age of grandparents and parents in the birth month of the child. I also construct a variable ranging from 1 to 13 that indicates the birth month of the child, where 1 indicates a birth in October 2001 and 13 a birth in October 2002.

Cohabitation. With data from the BRP, I construct variables indicating if both parents or both grandparents are registered at the same address in the child's birth month. The address data is limited to all officially registered addresses in the Netherlands.

Benefits. With data from tax records, I construct dummy variables equal to 1 when a parent or grandparent receives benefits in the month of birth. I classify a parent or grandparent as receiving benefits when they receive social welfare benefits, social benefits, unemployment benefits, or disability support.

College degree. With data from the Central Register of Higher Education Programmes, I obtain the highest completed degree for both parents in October 2001[9] and create a dummy variable equal to 1 when parents have at least a college degree[10].

Firstborn. With data on children and their registered parents, I construct a dummy variable equal to 1 when the child born in the recruitment period is the mother's firstborn child.

Male. With information from birth registries, I create a dummy variable equal to 1 when the child's registered sex at birth is male.

Outcomes

Employment. With monthly data from tax records, I construct variables that inform on

[7] Linkage failures are often due to inaccuracies in registered addresses or birth dates.

[8] In the administrative data available at the time of analysis, same-sex couples are classified as one mother and one father. I follow this classification in the analysis.

[9] Highest obtained education is recorded each year on October 1.

[10] College degree is defined as the Dutch hbo bachelor degree, SOI code 3111. Any degree with a higher SOI code, for example a university bachelor degree, is counted as at least a college degree.

employment status of parents and grandparents. I classify a parent or grandparent as employed in a given month if they receive income from employment as an employee, as a director, or as a self-employed worker. Whether an individual receives income as an employee or director is registered monthly. Information on self-employment income is derived from annual tax records. To construct monthly employment indicators, individuals are classified as self-employed in all months of a given fiscal year if they report any self-employment income in that year. With this information, I construct dummy variables equal to 1 when a parent or grandparent is employed in the month of birth. I also create variables that indicate the number of months a parent or grandparent is employed in the first, second, and third year after birth.

Channels

Pension. With information from tax records, I create dummy variables equal to 1 when the grandparent receives pension in the month of birth, 12 months after birth, and 24 months after birth. I classify a grandparent as receiving pension when they receive income from the state pension (AOW), the surviving relatives pension (Anw), other pensions, or a foreign pension in a given month. The state pension (AOW) is provided to individuals who have reached the statutory pension age (65 for the majority of grandparents in the sample) and who have lived or worked in the Netherlands between ages 15 and the pension age, with the amount of pension proportional to the total years of residence or work. The surviving relatives pension is for individuals below the state pension age whose partner has passed away. Other pensions might include pension payments due to early retirement.

3.4 Study Population

I compare the analysis sample to the broader screener population using administrative and survey data collected around the time of the child's birth. I split the screener sample, including parents linked to administrative data in families where at least one parent filled out an item of the IPARAN screener, into three groups. Group 1, the analysis sample, consists of families eligible due to maternal maltreatment who are randomised into the treatment or control group. Group 2, not participating, consists of families where the mother reported having experienced maltreatment, but families are not participating in the randomised experiment. Group 3, not maltreated, consists of families where the mother did not report having experienced maltreatment.

I show that parents in the analysis sample have a lower socio-economic status and are less likely to report social support than parents in families where the mother did not report maltreatment. I also show evidence of assortative mating by maltreatment history. Socio-economic gradients for grandparents are less pronounced.

Self-reported childhood maltreatment. Of the 4,834 mothers who completed the IPARAN screener, 11.1 percent report having been maltreated in their childhood. Of the 4,662 fathers who completed the IPARAN screener, 7.2 percent report having been maltreated in their childhood (Figure 1). For both parents, emotional maltreatment is the most common

form of maltreatment. Physical maltreatment is the least common form for mothers, and sexual maltreatment is the least reported form for fathers.

By construction, every mother in the analysis sample experienced childhood maltreatment. As in the screener sample, physical maltreatment is the least common form of maltreatment for these mothers. Of the fathers in the analysis sample, 18.6 percent reported childhood maltreatment, with emotional maltreatment and witnessing parental violence the most commonly reported forms among these fathers. The proportion of fathers having experienced maltreatment is substantially higher in the analysis sample than in the screener sample.

Socioeconomic status. Table 1 shows that mothers in the analysis sample are, on average, older ($\hat{\beta} = -1.007$, $p < 0.05$) and more likely to have a college degree ($\hat{\beta} = -0.092$, $p < 0.01$) than maltreated mothers who are not participating in the RCT. Relative to mothers who were not maltreated, mothers in the analysis sample are less likely to be employed ($\hat{\beta} = 0.145$, $p < 0.01$) and more likely to be on benefits ($\hat{\beta} = -0.104$, $p < 0.01$) in the child's month of birth. The pattern is similar for fathers, who are less likely to be employed ($\hat{\beta} = 0.033$, $p < 0.05$) and more likely to be on benefits ($\hat{\beta} = -0.032$, $p < 0.05$) than fathers whose partner was not maltreated. This shows that parents in families where the mother was maltreated are more socio-economically disadvantaged than parents in families where the mother did not experience maltreatment.

Social support. Table 1 shows that mothers in the analysis sample are less likely to report feeling accepted by their family and less likely to report sufficient general support than maltreated mothers who are not participating in the RCT ($\hat{\beta} = 0.109$ and $\hat{\beta} = 0.087$, both $p < 0.05$). They also report substantially lower family acceptance and less general support than mothers who were not maltreated ($\hat{\beta} = 0.431$ and $\hat{\beta} = 0.321$, both $p < 0.01$). Fathers show a similar, but less pronounced, gradient: fathers in the analysis sample report lower family acceptance ($\hat{\beta} = 0.093$, $p < 0.01$) and less general support ($\hat{\beta} = 0.252$, $p < 0.01$) than fathers whose partner was not maltreated. Families in the analysis sample therefore report weaker informal support networks than families where the mother was not maltreated, and, for mothers, also weaker than families where the mother was maltreated but did not participate in the RCT.

Cohabitation and assortative mating. Table 1 shows that mothers in the analysis sample are less likely to cohabit with the child's father than mothers who were not maltreated (6.2 percentage points, $p < 0.05$) in the child's month of birth.

Table 2 shows that mothers who were maltreated are more likely to have a child with a father who also experienced maltreatment than mothers who were not maltreated. Among families in the analysis sample, 18.6 percent of fathers report having experienced maltreatment themselves, compared with 18.7 percent for families where the mother reported maltreatment but who were not participating in the RCT, and only 5.9 percent for families where the mother was not maltreated. Fathers are also more likely to be unregistered as a parent when the mother reported maltreatment (3.7 and 5.0 percent for the analysis and not-participating groups, respectively) than when she did not (1.4 percent). This pattern is consistent with assortative mating on

childhood maltreatment. Mothers with a history of maltreatment are more likely to form relationships with partners who share that history.

Grandparents. Tables 3 and 4 extend the comparison to the children's grandparents.

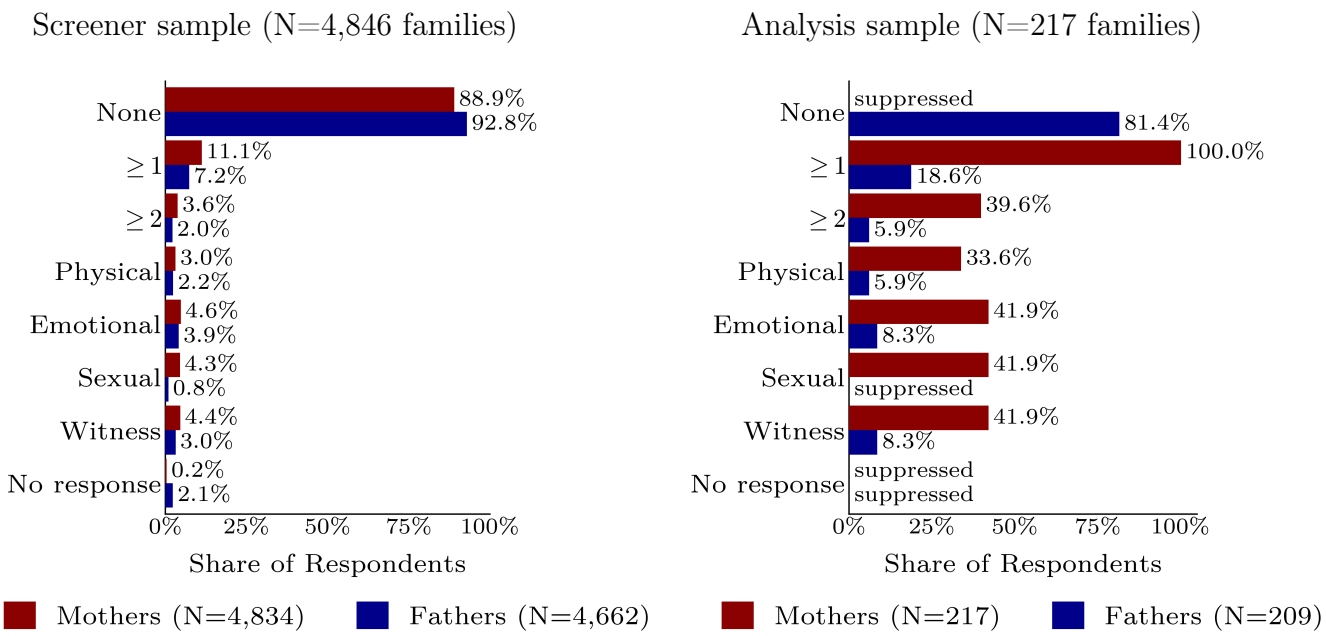
Grandmothers show a more modest socioeconomic gradient than parents. Grandmothers in the analysis sample are not statistically different from either comparison group in age, employment, or pension receipt, but are more likely to receive benefits than grandmothers in families where the mother did not report maltreatment ($\hat{\beta} = -0.055$ and $\hat{\beta} = -0.053$ for maternal and paternal grandmothers, respectively, both $p < 0.10$). Grandmothers in the analysis sample do not differ significantly from grandmothers in not participating families on any of these measures.

Maternal grandfathers in the analysis sample do not differ significantly from grandfathers in the not participating and not maltreated families. Paternal grandfathers in the analysis sample are less likely to be employed than paternal grandfathers in families where the mother did not report maltreatment ($\hat{\beta} = 0.063$, $p < 0.10$). Paternal grandfathers in the analysis sample are more likely to receive a pension than paternal grandfathers in not participating families ($\hat{\beta} = -0.109$, $p < 0.05$).

Grandparents in the analysis sample therefore look more similar to the broader screener population than parents in the analysis sample. However, it is important to take into account that baseline employment rates for grandparents are low, that pension status includes state pension receipt determined by age, and that there is less data on grandparents than on parents.

Representativeness. Van der Voort et al. (unpublished) report descriptive statistics for all parents in the Netherlands with a child born between October 1, 2001, and October 31, 2002, and for all parents in the catchment area with a child born in this period, regardless of whether they completed the IPARAN screener. Table D.5 reproduces and extends their summary statistics with additional descriptives. The table shows that parents in the screener population are more likely to be employed, have a college degree, and cohabit in the child's birth month compared with parents in the Netherlands and parents in the catchment area. This points to a selection bias among parents who completed the IPARAN screener, and it suggests that the conclusions drawn in this section may not generalise to all parents.

Figure 1: Self-reported Maltreatment



Notes: This figure shows the proportion of mothers and fathers that self-reported having experienced no child maltreatment, at least one type of child maltreatment, at least two types of child maltreatment, physical child maltreatment, emotional child maltreatment, sexual child maltreatment, being a witness to parental violence, or where the parent did not fill out the IPARAN screener. The proportions are displayed for parents in the screener sample and in the analysis sample. Note that categories with fewer than 10 responses are suppressed.

Table 1: Descriptive Statistics for Parents

	Mothers			Fathers		
	analysis	not participating	not maltreated	analysis	not participating	not maltreated
			A. Administrative data			
Age	31.924	−1.007**	−0.221	34.809	−0.361	−0.610
Employed	0.691	−0.040	0.145***	0.938	0.009	0.033**
Benefits	0.203	0.021	−0.104***	0.053	−0.023	−0.032**
College degree	0.221	−0.092***	0.021	0.206	−0.070**	0.014
Cohabitation	0.857	−0.040	0.062**			
Observations	217	318	4311	209	302	4251
			B. Survey data			
Family acceptance	0.421	0.109**	0.431***	0.721	0.004	0.093***
Observations	214	302	4247	201	280	4094
General support	0.373	0.087**	0.321***	0.468	0.090*	0.252***
Observations	217	315	4255	203	287	4122

Notes: Column 1 reports the analysis-sample mean for mothers and fathers, from a regression of each variable on group indicators with the analysis sample as the omitted category and heteroskedasticity-robust standard errors. Columns 2 and 3 report the difference from this mean for, respectively, mothers who were maltreated but not participating in the RCT and mothers who were not maltreated. Administrative data informs on parents' age, employment status, benefit receipt, college education, and cohabitation status in the child's birth month.

Survey data provide information on whether parents feel accepted by their family and perceive sufficient general support, measured within two weeks after birth. In accordance with the privacy regulations of Statistics Netherlands, potentially disclosive statistics are not reported. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2: Assortative Mating

	Fathers			
	not maltreated	maltreated	not registered	N
Analysis	0.777	0.186	0.037	217
Mothers not participating	0.762	0.187	0.050	318
Mothers not maltreated	0.928	0.059	0.014	4311

Notes: This table displays the proportion of fathers who reported not to have experienced maltreatment (column 1), to have experienced maltreatment (column 2), and who were not registered as a parent (column 3), for families in the analysis sample, families where the mother experienced maltreatment but did not participate in the RCT, and families where the mother did not report maltreatment.

Table 3: Descriptive Statistics for Grandmothers

	Mothers' mothers			Fathers' mothers		
	analysis	not participating	not maltreated	analysis	not participating	not maltreated
Age	58.592	−0.941	0.508	62.332	−0.624	−0.396
Employed	0.304	0.072	0.008	0.219	0.051	0.035
Pension	0.298	−0.054	−0.011	0.428	−0.067	−0.022
Benefits	0.171	0.061	-0.055^{*}	0.155	−0.037	-0.053^{*}
Cohabitation						
Observations	181	258	3953	187	255	3893

Notes: Column 1 reports the analysis-sample mean for maternal and paternal grandmothers, from a regression of each variable on group indicators with the analysis sample as the omitted category and heteroskedasticity-robust standard errors. Columns 2 and 3 report the difference from this mean for grandmothers whose child's mother was maltreated but not participating in the RCT and grandmothers whose child's mother was not maltreated. Administrative data informs on grandmothers' age, employment status, pension receipt, benefit receipt, and cohabitation status in the child's birth month. In accordance with the privacy regulations of Statistics Netherlands, potentially disclosive statistics are not reported. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Descriptive Statistics for Grandfathers

	Mothers' fathers			Fathers' fathers		
	analysis	not participating	not maltreated	analysis	not participating	not maltreated
Age	61.171	−0.785	0.288	64.960	−0.833	−0.890
Employed	0.465	0.019	0.045	0.349	0.060	0.063*
Pension	0.382	−0.070	0.046	0.583	−0.109**	−0.056
Benefits	0.229	0.037	−0.050	0.126	0.046	0.028
Cohabitation						
Observations	170	240	3778	175	245	3622

Notes: Column 1 reports the analysis-sample mean for maternal and paternal grandfathers, from a regression of each variable on group indicators with the analysis sample as the omitted category and heteroskedasticity-robust standard errors. Columns 2 and 3 report the difference from this mean for grandfathers whose child's mother was maltreated but not participating in the RCT and grandfathers whose child's mother was not maltreated. Administrative data informs on grandfathers' age, employment status, pension receipt, benefit receipt, and cohabitation status in the child's birth month. In accordance with the privacy regulations of Statistics Netherlands, potentially disclosive statistics are not reported. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.5 Empirical Framework

I estimate the effect of Supportive Parenting on grandparental childcare and employment, using the random assignment of families to treatment. Because assignment is random, treatment status T_i is orthogonal to potential outcomes, so a simple comparison of means across the treatment and control groups identifies the intention-to-treat (ITT) effect of being offered SP. I estimate this effect with the following OLS regression, run separately for each outcome period t:

$$Y_{it} = \beta_{0t} + \beta_{1t} T_i + \sum_j \beta_{jt} X_{ij} + \varepsilon_{it}, \tag{1}$$

where T_i equals 1 when a family is randomised into the treatment group and 0 otherwise, and X_{ij} is a set of j baseline control variables.[11] In eight families, there is no registered father. For these families, I set the paternal controls to 0 and add a dummy variable indicating whether the father of the child is registered. The periods t differ by outcome. For grandparental childcare, t indexes 6 weeks, 12 months, and 24 months after birth. For employment, t indexes the number of months a parent or grandparent is employed in the first (months 1–12), second (months 13–24), and third (months 25–36) year after birth.

Specifications. I report three specifications: (1) without X_{ij}, (2) with X_{ij}, and (3) with X_{ij} and non-random attrition weights. I add control variables to improve the precision of the estimates. Not all surveys used to construct outcomes, such as grandparental childcare, are completed for every family in every period. To correct for possible differential attrition

[11] X_{ij} includes age at birth, college degree, employment status, benefit receipt, and cohabitation for both parents at birth. It also includes the child's birth month, whether the child is the mother's firstborn, and male.

between the treatment and control groups, I use Hainmueller entropy weights (Hainmueller, 2012; Hainmueller and Xu, 2013). Among families with a non-missing outcome, the weights rebalance the control group so that the mean and variance of the covariates match those of the treatment group. I use the same set of X_{ij} to construct the entropy weights, except that I replace missing values of control variables with the mean (continuous variables) or the mode (binary variables) rather than setting them to 0 and adding a missingness dummy. In the robustness analysis, I report alternative specifications in which I do not impute missing values, in which I use an alternative set of X_{ij}, and in which I construct entropy weights using only the first moment of the covariates (Table E.10).

Multiple hypothesis testing. In the heterogeneity analyses, I test a large number of related hypotheses. Testing many hypotheses simultaneously increases the likelihood of finding at least one statistically significant result by chance alone, so evaluating each p-value against a conventional significance threshold without adjustment would overstate the evidence against the null. I address this using the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995), which controls the false discovery rate (FDR). Within each family of m related tests, I rank p-values from smallest to largest, $p_{(1)} \leq \cdots \leq p_{(m)}$, and compute the adjusted p-value at rank i as

$$q_{(i)} = \min\left(1,\ \min_{k \geq i}\ p_{(k)} \times \frac{m}{k}\right).$$

I reject hypotheses with $q_{(i)} \leq 0.10$. I apply this correction to the heterogeneity analyses and to the covariate balance checks within heterogeneity subgroups.

Balance. For the method to be valid, families randomised into the control and treatment groups should not differ significantly on average at baseline. Table 5 shows that mothers in the treatment group are on average 1.25 years younger at baseline than mothers in the control group. There are no other significant differences between the control and treatment groups in the sample used for the main causal analyses ($N = 217$).

Compliance. Take-up of the home visits is high. In the sample used for the main causal analyses ($N = 217$), 94 percent of families randomised into the treatment group for whom visit data are available completed the intervention, defined as having 4 or more home visits.[12] No families randomised into the control group received home visits. Figure D.4 shows the distribution of take-up. Because non-compliance is low, I do not report the local average treatment effect (LATE).

[12]Of the 106 families randomised into the treatment group, 98 received four or more visits, six received three or fewer, and for two families information on the number of visits was unavailable.

Table 5: Balancing Table

	Control Mean	Treatment - Control
	A. Mother	
Age at birth	32.535	−1.250**
	[4.266]	(0.607)
College degree	0.261	−0.082
		(0.056)
Employed	0.667	0.003
		(0.064)
Benefits	0.216	−0.028
		(0.055)
Cohabitation	≥ 0.9	−0.031
		(0.037)
Observations	111	106
	B. Father	
Age at birth	35.332	−1.071
	[5.815]	(0.754)
College degree	0.224	−0.038
		(0.056)
Employed	≥ 0.9	0.045
		(0.031)
Benefits	≤ 0.1	−0.026
		(0.031)
Observations	107	102
	C. Child	
Birth month	7.306	−0.118
	[3.933]	(0.522)
Firstborn	0.414	0.010
		(0.067)
Male	0.613	−0.047
		(0.067)
Observations	111	106

Notes: This table shows the baseline differences between families randomised into the control and treatment group for families eligible due to maternal maltreatment. Baseline variables on the mother (A), father (B), and child (C) are from administrative data estimated in the month of birth, except for college degree, which is estimated in October 2001. Standard deviations are displayed in square brackets, robust standard errors are displayed in round parentheses, *p<0.10, **p<0.05, ***p<0.01. Note that, in accordance with the privacy regulations of Statistics Netherlands, potentially disclosive statistics are not reported.

4 Results

I estimate the effects of Supportive Parenting (SP) on grandparental childcare and employment among families in which the mother experienced childhood maltreatment. Section 4.1 shows that SP increases childcare provided by paternal grandparents, with no corresponding effect

for maternal grandparents. Section 4.2 shows that SP increases the number of months fathers are employed, while I find no evidence of effects on the total number of months mothers or grandparents are employed.

4.1 Grandparental Childcare

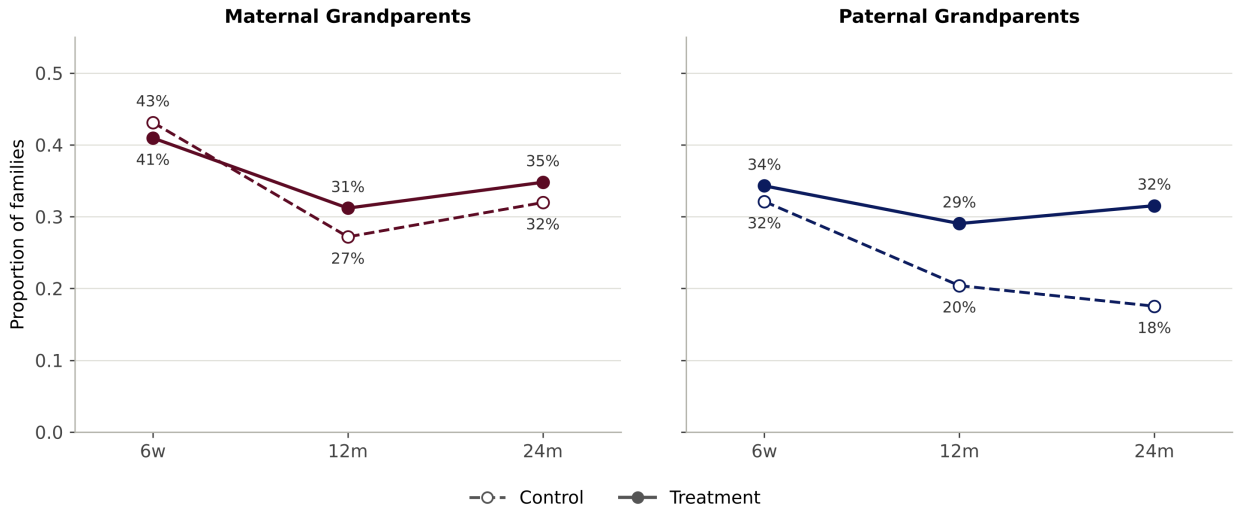
Among families eligible due to maternal maltreatment, 43 percent of control families and 41 percent of treatment families report receiving childcare from maternal grandparents at baseline. The corresponding shares for paternal grandparents are 32 percent and 34 percent. Figure 2a plots the proportion of families receiving grandparental childcare in the control and treatment groups, from maternal and paternal grandparents, at 6 weeks, 12 months, and 24 months.

SP increases childcare provided by paternal grandparents, with no evidence of an effect on childcare provided by maternal grandparents. Figure 2b reports the estimated intention to treat (ITT) effects and 95 percent confidence intervals under three specifications, namely (1) without covariates X_{ij}, (2) with X_{ij}, and (3) with X_{ij} and weights correcting for non-random attrition. At 12 months the point estimates range from 6.85 to 8.64 percentage points relative to a control group mean of 20 percent, though they are not statistically significant. By 24 months, the effect is significant across specifications, with estimates ranging from 11.87 to 15.22 percentage points against a control group mean of 18 percent.

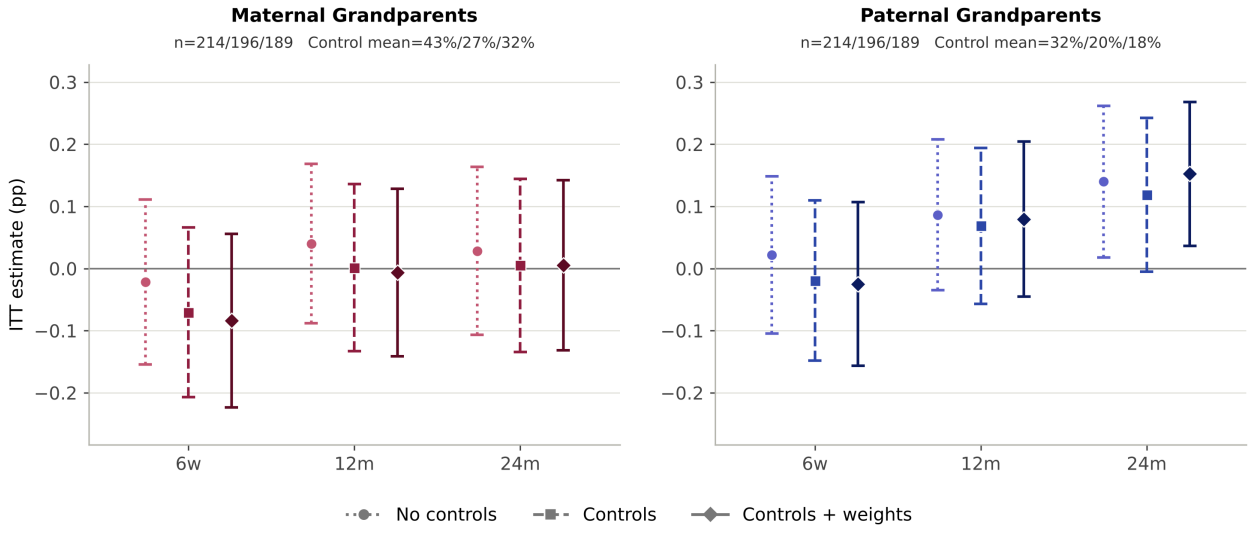
For maternal grandparents, I find no evidence of an effect on childcare provision. Across the treatment and control groups, 29 percent of families receive childcare from maternal grandparents at 12 months and 33 percent at 24 months, with treatment and control groups statistically indistinguishable throughout.

Figure 2: Grandparental Childcare

(a) Grandparental Childcare Proportions



(b) Grandparental Childcare Effects



Notes: Panel A shows the proportion of families receiving childcare from maternal and paternal grandparents at 6 weeks, 12 months, and 24 months, in families where the mother was maltreated as a child. Panel B shows the effect of Supportive Parenting on grandparental childcare at 6 weeks, 12 months, and 24 months, in families where the mother was maltreated as a child. The figure shows ITT estimates with 95 percent confidence intervals for regressions without X_{ij}, with X_{ij}, and with X_{ij} and non-random attrition weights. The point estimates can be found in Table D.6.

4.2 Employment

Childcare is time intensive, and an increase in childcare provided by others, for instance by grandparents, could relax time constraints on parents and allow them to increase labour supply

or return to work more quickly after birth (Bratti et al., 2018; Posadas and Vidal-Fernandez, 2013). At the same time, an increase in grandparental childcare may reduce grandparents' own labour supply (Backhaus and Barslund, 2021). To examine these dynamics, I analyse the number of months worked by parents and grandparents in the first, second, and third year after birth, where a month is counted as employed if the individual receives income from employment as an employee, director, or through self employment in that month. I find that SP increases the number of months worked by fathers, with no evidence of effects on the total number of months worked by mothers or grandparents.

Grandparents. Employment is low among paternal grandparents in the child's birth month (21.9 percent of grandmothers, 34.9 percent of grandfathers), with no significant differences between treatment and control. Over the first three years after birth, paternal grandmothers in the control group are employed for an average of 8.1 months and paternal grandfathers for 9.4 months, with no significant differences in total months employed (Table 6). Point estimates suggest a modest decrease in months worked for grandmothers and a modest increase for grandfathers, but only the estimate for grandfathers in the third year is statistically significant. Their employment increases by an estimated 1.4 months relative to a control group mean of 2.8 months.

Parents. In the child's birth month, 66.8 percent of mothers and 91.2 percent of fathers are employed. Over the first 36 months after birth, mothers in the control group are employed for an average of 24.5 months and fathers for 32.8 months. For mothers, point estimates suggest SP reduces employment in the first two years and increases it in the third, though none of these effects is statistically significant. For fathers, SP increases employment by an estimated 1.5 months over the first three years after birth, with the effect arising in the second and third years (0.6 months in each year). These results are reported in Table 6.

Table 6: Months Employed

	First Year		Second Year		Third Year		Total (Y1-Y3)		
	Control Mean	Treatment - Control	Control Mean	Treatment - Control	Control Mean	Treatment - Control	Control Mean	Treatment - Control	N
Mother	8.180	−0.091	8.054	−0.087	8.261	0.203	24.495	0.025	217
	[5.154]	(0.387)	[5.250]	(0.557)	[5.381]	(0.624)	[14.442]	(1.376)	
Father	11.131	0.274	10.860	0.590**	10.813	0.596*	32.804	1.459**	209
	[2.632]	(0.187)	[3.023]	(0.280)	[3.291]	(0.344)	[7.632]	(0.730)	
Paternal grandmother	2.871	−0.855	2.688	−0.699	2.495	−0.615	8.054	−2.169	187
	[5.119]	(0.735)	[4.963]	(0.740)	[4.731]	(0.723)	[14.579]	(2.174)	
Paternal grandfather	3.557	0.653	3.080	1.051	2.807	1.377*	9.443	3.081	175
	[5.384]	(0.802)	[5.059]	(0.749)	[5.042]	(0.781)	[14.952]	(2.187)	

Notes: This table shows the average number of months worked by mothers, fathers, paternal grandmothers, and paternal grandfathers, in the first year, second year, and third year after birth, in families where the mother was maltreated as a child. The last columns show the total number of months worked in the first three years after birth. For each period, the mean of the control group (Control Mean) and the estimated effect from a regression with X_{ij} and

non-random attrition weights (Treatment - Control) are displayed. Standard deviations are displayed in square brackets, robust standard errors are displayed in round parentheses, *p<0.10, **p<0.05, ***p<0.01.

4.3 Heterogeneity

To evaluate whether the programme's effects on grandparental childcare are concentrated among families with pre-existing supportive relationships, I analyse heterogeneity by baseline support. I find evidence that effects on paternal grandparental childcare are driven by families with higher baseline support. This suggests that nurses are effective in mobilising childcare among families with existing relationships and where mothers are able to ask for help. The results are also consistent with effects being concentrated among safe relationships, with no evidence of effects on maternal grandparental childcare, apart from the mothers who report always feeling accepted by their family at baseline.

I construct four indicators of baseline support: (1) general support, (2) family acceptance, (3) partner involvement, (4) ability to ask for help. All indicators are binary and divide the sample into families with lower (indicator = 0) and higher (indicator = 1) baseline support. Additionally, I create a support index that sums all indicators.

Figure 3 shows the ITT effect of Supportive Parenting on maternal and paternal grandparental childcare for the low and high baseline support groups in specifications with controls, and with controls and non-random attrition weights[13]. I report effects from the ITT specification with controls and non-random attrition weights in-text. For maternal grandparental childcare, only the families where the mother reports high baseline acceptance by her family show some evidence of an effect of SP, where the programme increases the probability of grandparents providing childcare at 24 months ($\hat{\beta} = 0.242$, $p = 0.053$). This shows that if SP increases maternal grandparental childcare, it is among the mothers who feel accepted by their family at baseline.

For paternal grandparental childcare, families with low baseline support show no evidence of effects for any of the indicators. In families with high baseline support, I find evidence that SP increases paternal grandparental childcare at 24 months. This holds for all indicators: general support ($\hat{\beta} = 0.489$, $p < 0.01$), family acceptance ($\hat{\beta} = 0.314$, $p < 0.01$), partner involvement ($\hat{\beta} = 0.197$, $p < 0.05$), and ability to ask for help ($\hat{\beta} = 0.307$, $p < 0.01$).

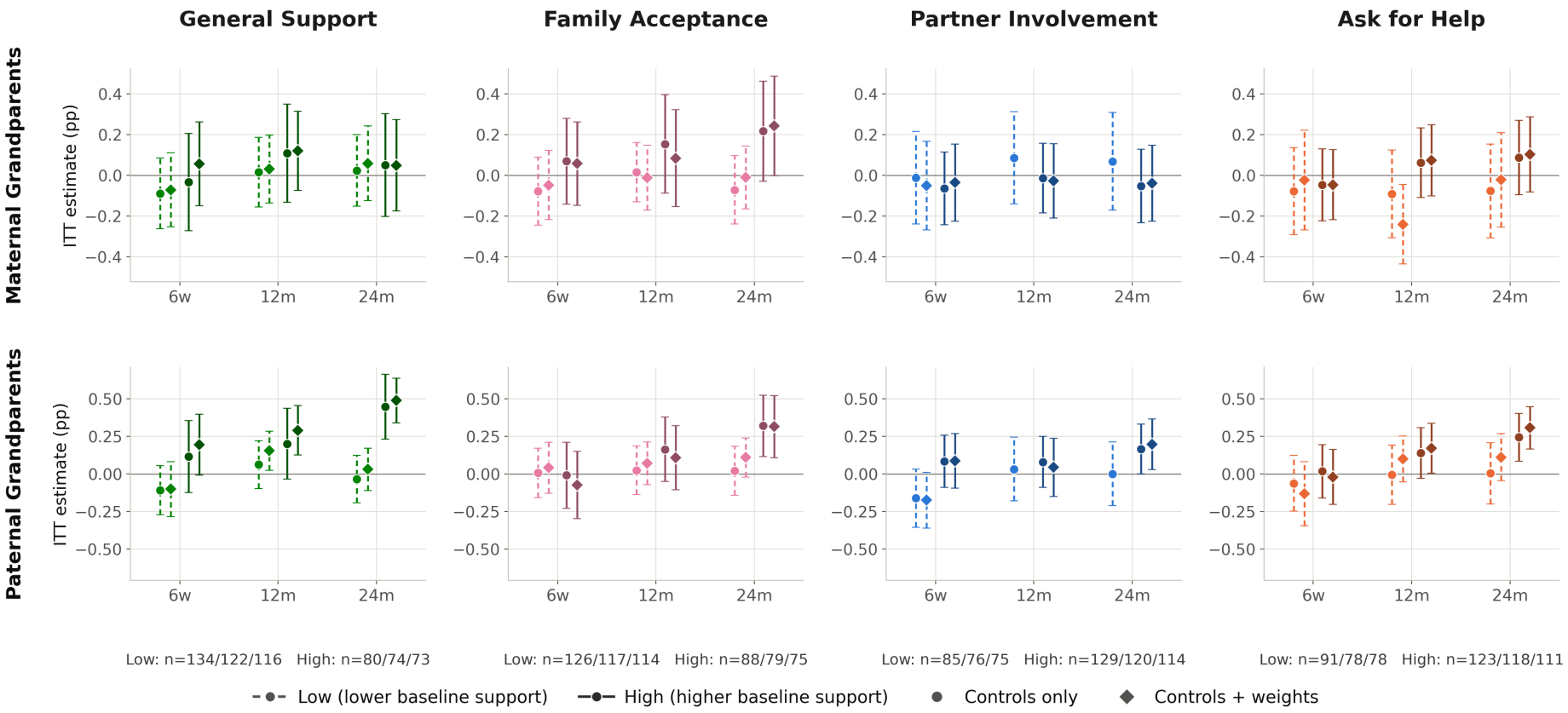
Table 7 shows whether treatment effects differ significantly by baseline support. The table shows the interaction effect of the baseline support indicator and randomisation in specifications with control variables and non-random attrition weights. For maternal grandparental childcare, there is no evidence of heterogeneous effects by baseline support at 24 months (all $p > 0.10$). For paternal grandparental childcare, treatment effects at 24 months are significantly higher for families with high baseline support. Three of the four indicators show evidence of heterogeneous effects at 24 months: general support ($\hat{\beta} = 0.467$, $p < 0.01$), family acceptance ($\hat{\beta} = 0.267$,

[13]Contrary to other analyses, I do not report effects without controls and weights, because there is some evidence of baseline imbalance for several of the heterogeneity checks.

$p < 0.05$), and ability to ask for help ($\hat{\beta} = 0.279$, $p < 0.05$). The support index shows the same pattern ($\hat{\beta} = 0.167$, $p < 0.01$). These findings are robust against Benjamini-Hochberg multiple hypothesis testing correction. In specifications with only controls, the interactions for paternal grandparental childcare at 24 months are of similar size and significance, although the family acceptance and ability to ask for help interactions no longer pass the Benjamini-Hochberg correction ($q = 0.112$). At 6 weeks (baseline), Figure 3 and Table 7 show one significant interaction, for partner involvement and paternal grandparental childcare ($\hat{\beta} = 0.275$, $p < 0.05$), and no evidence of differences for the other indicators. Partner involvement is also the only indicator with no evidence of a heterogeneous interaction effect at 24 months.

I also construct balance tables for all heterogeneity splits and find significant baseline differences for 11 variables, out of 96 tested, which is close to the 9.6 expected by chance under a 10% significance threshold (Table E.8). After correcting for multiple hypothesis testing (Benjamini-Hochberg, applied separately within each of the 8 heterogeneity subgroups, $K = 12$), I find 6 significant differences (Table E.9).

Figure 3: Heterogeneity by support



Notes: This figure shows the effect of Supportive Parenting on maternal and paternal grandparental childcare for families with low and high baseline support, according to four indicators: (1) general support, (2) family acceptance, (3) partner involvement, (4) ability to ask for help. The figure shows ITT effects and their 95% confidence intervals from specifications with controls, and with controls and non-random attrition weights. Note that for some analyses, only specifications with controls are displayed, because the algorithm used to rebalance does not converge on the specific sample.

Table 7: Interaction Effects by Baseline Support

	Maternal grandparents			Paternal grandparents		
	6 weeks	12 months	24 months	6 weeks	12 months	24 months
General	0.064	0.094	0.044	0.189	0.103	0.467***
	(0.149)	(0.142)	(0.147)	(0.147)	(0.130)	(0.119)
		[q=0.677]	[q=0.766]		[q=0.570]	[q=0.001]
Family	0.151	0.126	0.205	−0.045	0.070	0.267**
	(0.142)	(0.139)	(0.147)	(0.139)	(0.137)	(0.128)
		[q=0.677]	[q=0.469]		[q=0.694]	[q=0.099]
Partner	0.032	−0.066	−0.102	0.275**	−0.038	0.115
	(0.147)	(0.144)	(0.143)	(0.133)	(0.129)	(0.124)
		[q=0.737]	[q=0.677]		[q=0.769]	[q=0.566]
Help	0.045	0.263*	0.196	0.069	0.201	0.279**
	(0.142)	(0.139)	(0.145)	(0.135)	(0.133)	(0.126)
		[q=0.469]	[q=0.469]		[q=0.260]	[q=0.099]
Index§	0.049	0.070	0.051	0.079	0.063	0.167***
	(0.053)	(0.055)	(0.056)	(0.052)	(0.050)	(0.046)
		[q=0.282]	[q=0.357]		[q=0.282]	[q=0.001]
Observations	214	196	189	214	196	189

Notes: This table shows the interaction effect between randomisation into the treatment group and the baseline support indicators on childcare provided by maternal and paternal grandparents. The specification includes the general set of controls and non-random attrition weights, as well as controlling for randomisation and the support indicator or index. Standard errors in parentheses. q indicates the p-value adjusted for Benjamini-Hochberg multiple hypothesis correction. Corrections are applied separately for maternal and paternal grandparental childcare for the 8 interactions (4 support indicators at 12 and 24 months, $K = 8$ for each), and for the index at 12 and 24 months for maternal and paternal grandparental childcare ($K = 4$). Significance stars follow the unadjusted p-value: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

§Index = General + Family + Partner + Help

4.4 Channels

To analyse the mechanisms through which SP affects grandparental childcare, I analyse four possible channels: agreement on parenting, partner contribution, confiding about parenting difficulties, and grandparent pension receipt. I find no evidence to support that SP affects grandparental childcare through any of these channels. In-text point estimates refer to ITT specifications with controls and non-random attrition weights.

Agreement on Parenting

Intergenerational conflict, such as disagreement about parenting styles, is a common stressor between parents and grandparents (Bai et al., 2023; Tracy et al., 2022; Wright, 2023). During SP home visits, nurses help parents improve their parenting practices and reflect on their upbringing. This could contribute to the alignment between parents and grandparents in parenting approaches, potentially reducing conflict and facilitating grandparents' involvement in childcare. To explore this channel, I analyse whether parents report that their parents and in-laws agree with their childrearing practices. I find no evidence that SP affects grandparental childcare through increased agreement on parenting styles.

At 6 weeks, 12 months, and 24 months, one parent is asked whether their parents agree with their parenting approach. The parent is also asked whether their in-laws agree with their parenting approach. The questions are asked to all families, regardless of whether grandparents provide childcare. Among respondents at baseline, 78 percent of the control group families report that maternal grandparents always or often agree with their parenting approach and 70 percent report that paternal grandparents agree with their parenting approach. These shares do not differ significantly from those in the treatment group ($\hat{\beta} = -0.007$, $p = 0.91$ for maternal grandparents; $\hat{\beta} = 0.058$, $p = 0.50$ for paternal grandparents). At 12 and 24 months, there is no evidence that SP affects agreement on parenting style, for either maternal grandparents ($\hat{\beta} = -0.041$, $p = 0.58$ at 12 months; $\hat{\beta} = -0.039$, $p = 0.61$ at 24 months) or paternal grandparents ($\hat{\beta} = -0.062$, $p = 0.36$ at 12 months; $\hat{\beta} = 0.033$, $p = 0.67$ at 24 months). The findings are displayed in Figure 4.

Partner Contribution

SP may affect the allocation of childcare and household tasks within the family. If grandparental childcare is a substitute for parental childcare, a reduction in parental childcare could increase grandparental childcare (Meil and Rogero-García, 2015). To explore this channel, I analyse whether parents indicate that their partner helps with childcare and whether their partner helps with household tasks. I find no evidence that SP increases paternal grandparental childcare through reduced partner contribution.

At 6 weeks, 12 months, and 24 months, one parent is asked whether their partner helps with childcare. The parent is also asked whether their partner helps with household tasks. The questions are asked to all families, regardless of whether grandparents provide childcare. Among respondents at baseline, 96 percent of the control group families report that their partner helps with childcare and 95 percent report that their partner helps with household tasks. These shares do not differ significantly from those in the treatment group ($\hat{\beta} = -0.028$, $p = 0.27$ for childcare; $\hat{\beta} = -0.050$, $p = 0.25$ for household tasks). At 12 and 24 months, there is no evidence that SP affects whether the partner helps with childcare ($\hat{\beta} = 0.006$, $p = 0.77$ at 12 months; $\hat{\beta} = -0.002$, $p = 0.95$ at 24 months). For household tasks, SP increases the probability that the partner helps at 12 months ($\hat{\beta} = 0.080$, $p = 0.075$), with no evidence of an effect at 24 months ($\hat{\beta} = 0.060$, $p = 0.25$). This effect goes in the opposite direction of the proposed

channel, which would require a reduction in partner contribution. The findings are displayed in Figure 4.

Although I find no evidence of a reduction in partner contribution, it is important to note that the proportion of parents reporting partner contribution in the control group is high in each time period, leaving little scope for change.

Confiding About Parenting Difficulties

SP nurses may encourage parents to confide more in others about parenting problems. This could be an indicator of making new connections or finding new sources of support. To explore this channel, I analyse whether parents indicate that they can talk to their partner when their child is difficult and whether they can talk to their friends or family when their child is difficult. I find no evidence that SP increases grandparental childcare through increased confiding about parenting difficulties.

At 6 weeks, 12 months, and 24 months, one parent is asked whether they can talk to their partner when their child is difficult. The parent is also asked whether they can talk to a friend or family member when their child is difficult. Among respondents at baseline, 98 percent of the control group families report that they can talk to their partner when their child is difficult and 98 percent report that they can talk to friends or family. These shares do not differ significantly from those in the treatment group ($\hat{\beta} = -0.018$, $p = 0.34$ for partner; $\hat{\beta} = -0.029$, $p = 0.22$ for friends/family). At 12 and 24 months, there is no evidence that SP affects confiding about parenting difficulties, for either partner ($\hat{\beta} = 0.009$, $p = 0.55$ at 12 months; $\hat{\beta} = -0.012$, $p = 0.72$ at 24 months) or friends/family ($\hat{\beta} = 0.025$, $p = 0.56$ at 12 months; $\hat{\beta} = -0.010$, $p = 0.72$ at 24 months). The findings are displayed in Figure 4.

Although I find no evidence of effects on confiding about parenting difficulties, it is important to note that the proportion of parents in the control group who report having someone to confide in is high in each time period, leaving little scope for improvement.

Grandparent Pension Receipt

SP nurses could encourage grandparents to provide more childcare, leading them to go on early pension. Grandparents' pension status is a positive predictor of childcare supply (Hamilton et al., 2025) and pension reforms show that retirement can increase childcare provision (Bratti et al., 2018; Kaufmann et al., 2022). To explore grandparents' availability as a channel, I analyse paternal grandparents' pension status. I find no evidence to support this channel.

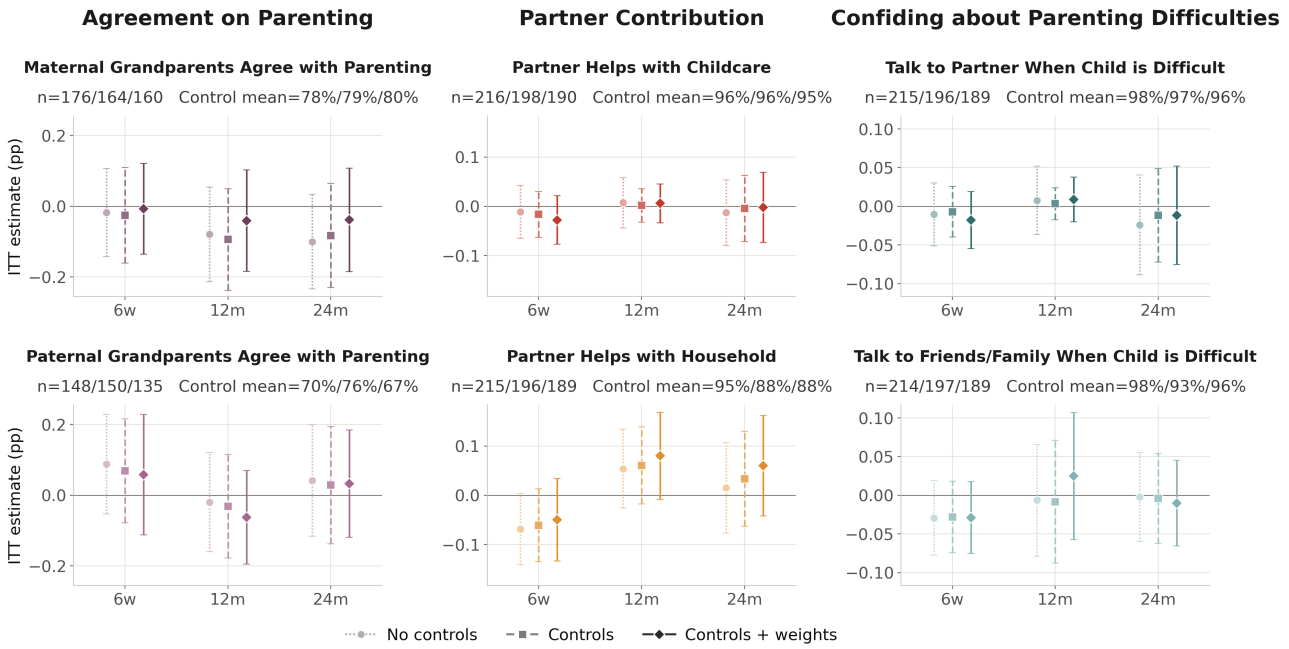
Pension status can be reached for endogenous reasons, such as early retirement. However, pension status can also be reached for exogenous reasons, namely by reaching the statutory pension age. I cannot distinguish between the two in the data.

At baseline, 49 percent of paternal grandmothers and 60 percent of paternal grandfathers in the control group are receiving a pension. This share does not differ significantly from the treatment group ($\hat{\beta} = -0.048$, $p = 0.48$ for grandmothers; $\hat{\beta} = 0.061$, $p = 0.41$ for grandfathers). Of the grandparents receiving a pension at baseline, only 18 percent of paternal

grandmothers and 27 percent of paternal grandfathers have not reached the statutory pension age, suggesting that pension status is mostly determined by reaching the statutory pension age rather than by early retirement. I find no evidence that SP affects whether paternal grandparents receive a pension 12 and 24 months after birth (grandmothers: $\hat{\beta} = -0.023$, $p = 0.72$ at 12 months, $\hat{\beta} = 0.021$, $p = 0.74$ at 24 months; grandfathers: $\hat{\beta} = 0.059$, $p = 0.44$ at 12 months, $\hat{\beta} = 0.024$, $p = 0.76$ at 24 months). The findings are displayed in Figure 5.

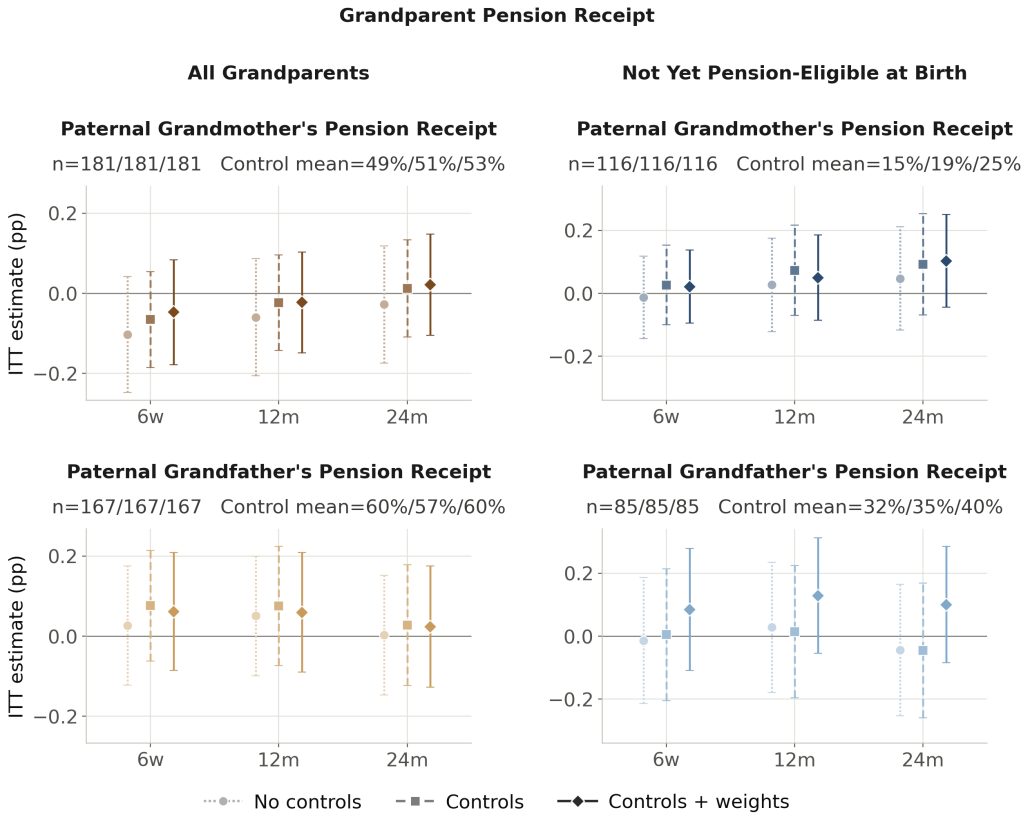
These findings are robust against restricting the sample to grandparents who had not yet reached the statutory pension age in the child's month of birth. In this subsample, 15 percent of paternal grandmothers and 32 percent of paternal grandfathers in the control group are receiving a pension at baseline, rising to 25 percent and 40 percent, respectively, by 24 months as more grandparents cross the statutory pension age. I find no evidence that SP affects pension receipt among these grandparents, at 12 months ($\hat{\beta} = 0.050$, $p = 0.47$ for grandmothers; $\hat{\beta} = 0.128$, $p = 0.17$ for grandfathers) or at 24 months ($\hat{\beta} = 0.102$, $p = 0.17$ for grandmothers; $\hat{\beta} = 0.100$, $p = 0.29$ for grandfathers). The findings for this restricted sample are also displayed in Figure 5.

Figure 4: Channels: Agreement, Partner Contribution, and Confiding



Notes: This figure shows the effect of Supportive Parenting on three potential channels for grandparental childcare: (1) agreement on parenting, (2) partner contribution, (3) confiding about parenting difficulties. The figure shows ITT effects and their 95% confidence intervals from specifications without controls, with controls, and with controls and non-random attrition weights.

Figure 5: Channels: Grandparent Pension Receipt



Notes: This figure shows the effect of Supportive Parenting on grandparent pension receipt, for paternal grandmothers and paternal grandfathers, both for the full sample and for the subsample of grandparents who had not yet reached the statutory pension age at the child's birth. The figure shows ITT effects and their 95% confidence intervals from specifications without controls, with controls, and with controls and non-random attrition weights.

4.5 Robustness

In Appendix E, I show that my findings are not driven by substitution, choice of control variables and weights, paternal childhood maltreatment, or father's survey response. I summarise my findings here.

Substitution

To ensure that the increase in grandparental childcare is not a substitution for other support, I analyse the effect of SP on other self-reported support. I find no evidence that SP affects the likelihood of being in contact with other parents, friends, or neighbours. These findings can be found in Figure E.5.

Controls and weights

To show that my findings are robust against changes in the set of control variables and non-random attrition weight specifications, I run my analyses using alternative specifications. First,

I include only maternal characteristics as control variables. Second, rather than imputing missing values with the mean (for continuous variables) or the mode (for binary variables), or replacing them by zero, I retain missing values as such, which reduces the analysis sample. Third, instead of balancing the non-random attrition weights on both the mean and variance of the control variables, I balance them only on the mean. My findings are robust against these changes in control variables and weight specifications. These analyses can be found in Table E.10.

Paternal childhood maltreatment

In my main analyses, I analyse the effect of SP on families where the mother self-reported childhood maltreatment, regardless of fathers' childhood experiences. In Figure E.6 I show that findings are similar when restricting the sample to families where mothers experienced maltreatment and fathers do not report maltreatment.

Father's survey response

With the exception of the IPARAN screener, all surveys were completed by one parent and the mother was the respondent in most cases (92–96% depending on the survey). In my main analyses, I include all survey responses. I show that my findings are similar when only using maternal responses. These analyses can be found in Figure E.6.

4.6 Limitations

In this section, I evaluate three limitations of the study. First, I address the potential for reporting bias inherent in the self-reported primary outcome. Second, I discuss how sample size constraints limit the scope of the analyses. Third, I elaborate on the limitations of the employment data.

Self-reports

Grandparental childcare is assessed through self-reports, which may be subject to reporting bias. Parents could report socially desirable answers, possibly to show improvement to the nurse. The programme could also affect how parents perceive and interpret support, rather than affecting the extent to which grandparents are involved in childcare. Lastly, reporting could be selective, possibly leading to non-random attrition bias. To address these concerns, I give arguments to support a change in grandparental childcare involvement rather than perception, and I show that there is no evidence for non-random attrition.

First, if the home visits would change parents' perception and interpretation of support, rather than an actual change in childcare provision, we would expect similar effects on other types of support. As discussed in Section 4.5, I find no evidence of effects of SP on other types of support (Figure E.5).

Second, I show that there is no evidence for non-random attrition. Table F.11 shows that baseline characteristics are balanced when conditioning on having non-missing data for paternal grandparental childcare at 6 weeks, 12 months, and 24 months, with the exception of maternal age, which also differs between the treatment and control group in the full analysis sample. The number of parents who do not report paternal grandparental childcare at 12 and 24 months is not significantly different between the control and treatment group (Table F.11). My in-text analyses are also robust against using non-random attrition weights. To construct these weights, I only use administrative data, not self-reported data.

Sample size

The sample size limits me from detecting small effects and analysing rare outcomes. For this reason, I do not examine additional outcomes that could capture increased support from grandparents, such as financial transfers and residential proximity. The sample size also limits me from analysing the effects of SP in families eligible due to paternal childhood maltreatment, since the sample is much smaller (N=113). Power calculations supporting these decisions can be found in Appendix G.

Employment data

The employment measure captures only whether parents receive income from work in a given month, not how many hours they work. This is especially relevant for mothers, since hours worked is the margin on which Dutch mothers usually adjust their labour supply (Begall and Grunow, 2015). Data on hours worked are unavailable in the administrative data for the period studied. The absence of effects on the extensive margin does not rule out effects on the intensive margin.

5 Conclusion

Mothers who have experienced maltreatment in their childhood are more likely to have strained intergenerational relationships, potentially limiting grandparental childcare support. Grandparental childcare involvement is important, as it can alleviate parental time constraints and increase labour market participation.

In this paper, I document that mothers who experienced maltreatment in their childhood have a lower socio-economic status and report less support. I show that an early-childhood intervention, Supportive Parenting, increases paternal grandparental childcare in families where the mother experienced maltreatment. Supportive Parenting consists of six home visits by a trained nurse who aims to prevent parenting problems. During the home visits, which take place in the first 18 months after birth, the nurse reflects with parents on childhood experiences and aims to increase their social support.

By linking RCT data to administrative data, I show that Supportive Parenting increases childcare provided by paternal grandparents. SP also increases the number of months fathers

are employed, with no evidence of effects on the total number of months mothers or grandparents are employed. The increase in grandparental childcare is driven by families with higher baseline support, which is consistent with nurses being effective in helping parents mobilise latent support rather than creating relationships among parents who are unable to ask for help.

I analyse four channels and find no evidence that SP affects grandparental childcare through these channels: agreement on parenting, partner contribution, confiding about parenting difficulties, and grandparent pension receipt. The findings do not rule out that the nurses reduce coordination costs by guiding parents step-by-step in increasing support or that the content of the visits increases parents' ability to identify whom to ask for childcare.

In this paper, I show that with just six nurse home visits, SP can increase grandparental childcare in families where mothers experienced maltreatment as a child—a group with low baseline support.

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A Maps

Figure A.1: Target Region



Notes: This map shows the municipalities participating in the SP experiment.

Figure A.2: Overview 2026



Notes: This map shows the municipalities offering SP in 2026.

B Eligibility Screener

To determine eligibility, both parents are asked to fill out 16 IPARAN items, of which the translation by van der Put et al. (2017) can be found in table B.1. Scores of both parents are calculated as displayed in table B.1 and add up to a family score. When a family score adds up to 1 or more, the family is eligible for Supportive Parenting. When one parent reports having experienced child maltreatment, the family gets a score of 1 and is automatically eligible.

Despite eligibility according to the screener, families are unable to participate if they speak insufficient Dutch or if they intent to move outside the catchment area of the intervention.

Table B.1: Screener items and scores

Item	Responses (scores)
1. I expect that I will receive enough support from my partner in the care of our child(ren)	1. always (0), 2. often (0), 3. sometimes (.5), 4. never (.5)
2. I am very worried about raising this child	1. always (0.5), 2. often (0.5), 3. sometimes (0), 4. never (0)
3. During the last nine months I have felt unhappy about expecting a baby	1. always (0.5), 2. often (0.5), 3. sometimes (0), 4. never (0)
4. As a child I generally felt safe in the vicinity of (one of) my parents/carers	1. always (0), 2. often (0), 3. sometimes (1), 4. never (1)
5. I feel that the relationship between my parents/carers was often violent when I was a child	1. yes (1), 2. no (0)
6. I feel that my parents/carers hit me too much as a child	1. yes (1), 2. no (0)
7. Looking back at the last three years, I suffered serious mental problems (for a while)	1. yes (1), 2. no (0)
8. When I have/had an argument with my partner, we sometimes come/came to blows	Yes (1), No (0)
9. I should actually drink less alcohol/take fewer drugs	1. yes (1), 2. no (0)
10. A good upbringing involves smacking children regularly	1. yes (1), 2. no (0)
11. I was forcefully subjected to an unpleasant sexual experience before I was 16	1. yes (1), 2. no (0)
12. I feel very accepted by my family	1. always (0), 2. often (0.1), 3. sometimes (0.2), 4. never (0.3)
13. I can get so angry that I lose control	1. always (1), 2. often (1), 3. sometimes (0), 4. never (0)
14. I feel very much at home in the area where I live	1. always (0), 2. often (0.1), 3. sometimes (0.2), 4. never (0.3)
15. I would like to have more people around me on whom I could fall back for support	1. always (0.3), 2. often (0.2), 3. sometimes (0.1), 4. never (0.0)
16. I find it easy to ask for help or advice when I need it	1. always (0), 2. often (0.1), 3. sometimes (0.2), 4. never (0.3)

Notes: This table includes the IPARAN items used to determine eligibility for Supportive Parenting.

C Data

Figure C.3: Timeline Visits and Surveys

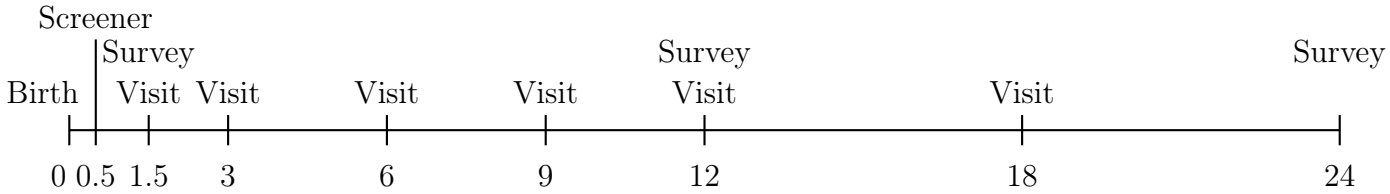


Table C.2: Survey Data

Variables	Time	Survey	Items	Definition
			Sample identification	
Parents maltreated during their childhood	14 days	IPARAN Screener	1.) I feel that my parents/carers hit me too much as a child. 2.) As a child I generally felt safe in the vicinity of (one of) my parents/carers. 3.) I was forcefully subjected to an unpleasant sexual experience before I was 16. 4.) I feel that the relationship between my parents/carers was often violent when I was a child.	The maternal maltreatment variable equals 1 if the mother reports maltreatment on one or more of the four items (yes to items 1, 3, or 4; sometimes or never to item 2). The paternal maltreatment variable is defined analogously for the father.
			Descriptive and heterogeneity	
General support	14 days	IPARAN Screener	1.) I would like to have more people around me on whom I could fall back for support.	The support variable equals 1 when the parent indicates never and 0 otherwise.
Family acceptance	14 days	IPARAN Screener	1.) I feel very accepted by my family.	The acceptance variable equals 1 when the parent indicates always and 0 otherwise.
Partner involvement	14 days	IPARAN Screener	1.) I expect to receive enough support from my partner in caring for my child(ren).	Partner involvement equals 1 when the mother indicates that she always expects to receive enough support from her partner and 0 otherwise.

Ability to ask for help	14 days	IPARAN Screener	1.) It is easy for me to ask for help or advice when I need it.	Ability to ask for help equals 1 when the parent indicates to always or often find it easy to ask for help and 0 otherwise.
		Baseline and outcomes		
Grandparental childcare	6 weeks, 12 months, 24 months	Support Survey	1.) Do you get help from your parents with raising your child? 2.) Do you get help from your in-laws with raising your child?	The maternal grandparental childcare variables equal 1 if the mother answers yes to the first question or if the father answers yes to the second question. The paternal grandparental childcare variables equal 1 if the father answers yes to the first question or if the mother answers yes to the second question. The 6-week measure serves as baseline.
		Channels		
Agreement on parenting	6 weeks, 12 months, 24 months	Support Survey	1.) Do your parents agree with your parenting approach? 2.) Do your parents-in-law agree with your parenting approach?	The maternal and paternal agreement variables equal 1 when the parent indicates that the respective grandparents always or often agree with their parenting approach.
Partner contribution	6 weeks, 12 months, 24 months	Support Survey	1.) Does your partner help with caring for your child? 2.) Does your partner help with household tasks?	The partner contribution variables equal 1 when the parent indicates that their partner helps with childcare and household tasks, respectively.
Confiding about parenting difficulties	6 weeks, 12 months, 24 months	Support Survey	1.) Are you able to talk to your partner when your child is being difficult? 2.) Are you able to talk to a friend or family member when your child is being difficult?	The confiding variables equal 1 when the parent indicates that they can talk to their partner and to a friend or family member, respectively, when the child is being difficult.
		Robustness		

Support from others	6 weeks, 12 months, 24 months	Support Survey	1.) Are you in contact with other parents your age? 2.) Are you in contact with your neighbours? If yes, do you get support from them? 3.) Are you in contact with friends or acquaintances?	For each type of support I create dummy variables equal to 1 if the parent indicates that they are in contact with or get support from that source.

D Additional Tables

Support

Table D.3: Baseline support

	Analysis	No maternal maltreatment	No parental maltreatment
Support mother	.37	.45	.43
Support father	.47	.47	.49
Family acceptance mother	.42	.68	.67
Family acceptance father	.72	.60	.68
Childcare maternal grandparents	.42	.56	.58
Childcare paternal grandparents	.33	.40	.41

Notes: This table shows the proportion of families where the mother reports feeling sufficiently supported within 14 days after birth, where the father reports feeling sufficiently supported within 14 days after birth, where the mother reports feeling accepted by her family within 14 days after birth, where the father reports feeling accepted by his family within 14 days after birth, where maternal grandparents provide childcare when the child is 6 weeks old, and where paternal grandparents provide childcare when the child is 6 weeks old. Proportions are displayed for families in the analysis sample, for families in the RCT where the mother did not report having experienced maltreatment, and for families in the RCT where neither parent experienced maltreatment.

Linkage

Table D.4: Children linked to family members

	Screener	Analysis
Children	4846	217
Mothers	4846	217
Fathers	4762	209
Maternal grandmothers	4392	181
Maternal grandfathers	4188	170
Paternal grandmothers	4335	187
Paternal grandfathers	4042	175

Notes: This table shows how many children could be linked to their mother, father, maternal grandmother, maternal grandfather, paternal grandmother, and paternal grandfather. These numbers are shown for the screener sample, used for descriptive statistics, and for the analysis sample, used for causal analysis.

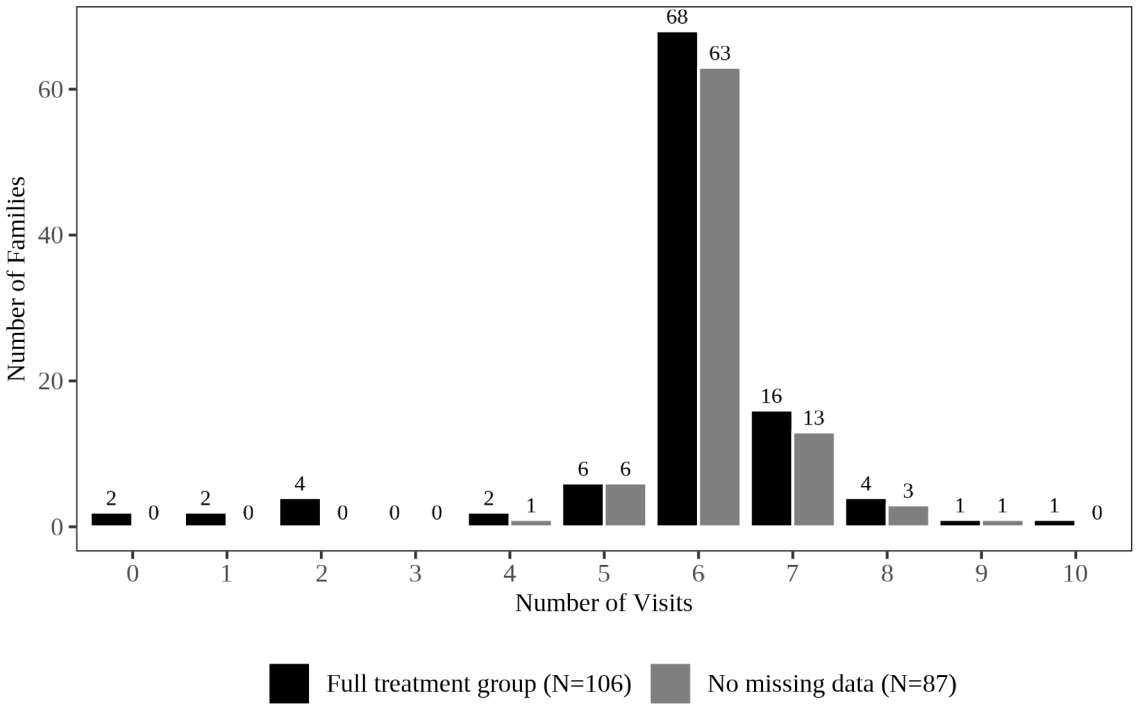
Descriptives

Table D.5: Descriptive Statistics for Parents

	Mothers			Fathers		
	Netherlands	catchment	screener	Netherlands	catchment	screener
Age	30.673	31.500	31.661	33.718	34.343	34.241
Employed	0.660	0.759	0.809	0.878	0.948	0.970
Benefits			0.112			0.023
College degree	0.159	0.195	0.234	0.160	0.200	0.214
Cohabitation	0.851	0.892	0.949			
Observations	239,372	8,799	4,846	228,472	8,593	4,762

Notes: This table shows descriptive statistics for parents with a child born in the recruitment period in the Netherlands, in the catchment area, and in families where one or both parents answered maltreatment questions as part of the IPARAN screener (screener). Descriptives on age, employment, college degree, and cohabitation for parents in the Netherlands and in the catchment area come from Van der Voort et al. (unpublished). Benefit receipt is not reported by Van der Voort et al. (unpublished) and is therefore shown for the screener sample only. The catchment-area sample contains 8,805 families. In 6 of these families no mother is registered, so the number of mothers (8,799) is lower than the number of families.

Figure D.4: Distribution Home Visits



Notes: This figure displays the distribution of home visits for two groups: (1) the full treatment group consisting of families randomised into the treatment group eligible due to maternal maltreatment, and (2) the non-missing data group consisting of families randomised into the treatment group eligible due to maternal maltreatment with complete data for all paternal grandparental childcare variables.

Table D.6: Grandparental Childcare: Estimates Behind Figure 2

		Control mean	Treatment mean	Without X_{ij} (1)	With X_{ij} (2)	With X_{ij} and weights (3)	N
Maternal grandparents	6 weeks	0.431	0.410	−0.022	−0.071	−0.084	214
				(0.068)	(0.070)	(0.071)	
				[0.750]	[0.313]	[0.239]	
	12 months	0.272	0.312	0.040	0.001	−0.006	196
				(0.065)	(0.068)	(0.069)	
				[0.541]	[0.984]	[0.928]	
	24 months	0.320	0.348	0.028	0.005	0.005	189
				(0.069)	(0.071)	(0.070)	
				[0.683]	[0.943]	[0.939]	
Paternal grandparents	6 weeks	0.321	0.343	0.022	−0.019	−0.025	214
				(0.065)	(0.066)	(0.067)	
				[0.737]	[0.769]	[0.710]	
	12 months	0.204	0.290	0.086	0.068	0.079	196
				(0.062)	(0.064)	(0.064)	
				[0.164]	[0.287]	[0.215]	
	24 months	0.175	0.315	0.140**	0.119*	0.152**	189
				(0.062)	(0.063)	(0.059)	
				[0.026]	[0.062]	[0.011]	

Notes: This table shows the ITT effect of Supportive Parenting on childcare provided by maternal and paternal grandparents at 6 weeks, 12 months, and 24 months, in families where the mother was maltreated as a child. Column (1) is estimated without X_{ij}, column (2) with X_{ij}, and column (3) with X_{ij} and non-random attrition weights. Control mean is the mean of the control group and treatment mean is the control mean plus the estimate in column (1). Robust standard errors in parentheses and p-values in square brackets. The 6-week survey is completed before the first home visit and serves as a baseline. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Robustness

Heterogeneity

Table E.7: Treatment × baseline social support: interaction effects (controls), BH-adjusted

	Maternal grandparents			Paternal grandparents		
	6 weeks	12 months	24 months	6 weeks	12 months	24 months
General	0.033	0.041	0.072	0.205	0.094	0.434***
	(0.147)	(0.143)	(0.149)	(0.142)	(0.133)	(0.125)
		[q=0.886]	[q=0.883]		[q=0.548]	[q=0.004]
Family	0.135	0.152	0.309**	0.011	0.177	0.284**
	(0.138)	(0.137)	(0.147)	(0.136)	(0.137)	(0.135)
		[q=0.537]	[q=0.287]		[q=0.314]	[q=0.112]
Partner	0.001	−0.016	−0.064	0.254**	0.034	0.128
	(0.143)	(0.139)	(0.146)	(0.129)	(0.134)	(0.133)
		[q=0.908]	[q=0.883]		[q=0.800]	[q=0.450]
Help	0.110	0.216	0.239*	0.108	0.191	0.264**
	(0.139)	(0.138)	(0.144)	(0.128)	(0.130)	(0.130)
		[q=0.314]	[q=0.314]		[q=0.281]	[q=0.112]
Index§	0.048	0.067	0.082	0.091	0.085	0.164***
	(0.052)	(0.054)	(0.058)	(0.051)	(0.052)	(0.050)
		[q=0.218]	[q=0.208]		[q=0.202]	[q=0.004]
Observations	214	196	189	214	196	189

Notes: This table shows the interaction effect between randomisation into the treatment group and the baseline support indicators on childcare provided by maternal and paternal grandparents. The specification includes the general set of controls, as well as controlling for randomisation and the support indicator or index. Standard errors in parentheses. q indicates the p-value adjusted for Benjamini-Hochberg multiple hypothesis correction. Corrections are applied separately for maternal and paternal grandparental childcare for the 8 interactions (4 support indicators at 12 and 24 months, $K = 8$ for each), and for the index at 12 and 24 months for maternal and paternal grandparental childcare ($K = 4$). Significance stars follow the unadjusted p-value: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

§Index = General + Family + Partner + Help

Table E.8: Covariate balance within heterogeneity subgroups

Heterogeneity	Low or high	Significant baseline differences		N
		$p < 0.10$	$q < 0.10$	
General support	low	3 / 12	2 / 12	292
General support	high	3 / 12	1 / 12	208
Family acceptance	low	1 / 12	0 / 12	221
Family acceptance	high	0 / 12	0 / 12	279
Partner involvement	low	1 / 12	1 / 12	202
Partner involvement	high	1 / 12	1 / 12	298
Ask for help	low	1 / 12	1 / 12	215
Ask for help	high	1 / 12	0 / 12	285

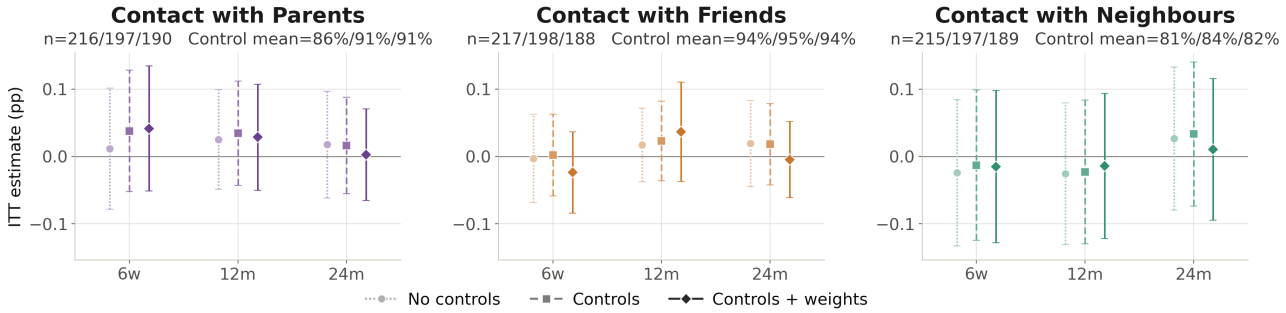
Notes: For each heterogeneity subgroup, randomisation is regressed on the 12 baseline controls used in the main balance table. Each column reports the number of covariates that are significantly different from zero, according to the standard p-values (p) and the Benjamini-Hochberg correction (q).

Table E.9: Covariates imbalance after Benjamini-Hochberg correction

Heterogeneity	Low or high	Covariate	Coef.	p	q
Partner involvement	low	Paternal employment	0.646	0.0000	0.000
Partner involvement	high	Cohabitation	−0.493	0.0000	0.000
Ask for help	low	Paternal employment	0.506	0.0001	0.002
General support	low	Maternal age	−0.029	0.0123	0.099
General support	low	Maternal employment	0.257	0.0166	0.099
General support	high	Maternal employment	−0.393	0.0050	0.060

Notes: Covariates with $q < 0.10$ (Benjamini-Hochberg corrected within each subgroup, $K = 12$) from regressions of randomisation on the 12 baseline controls, run separately within each of the 8 heterogeneity subgroups (4 moderators × 2 values).

Figure E.5: Substitution



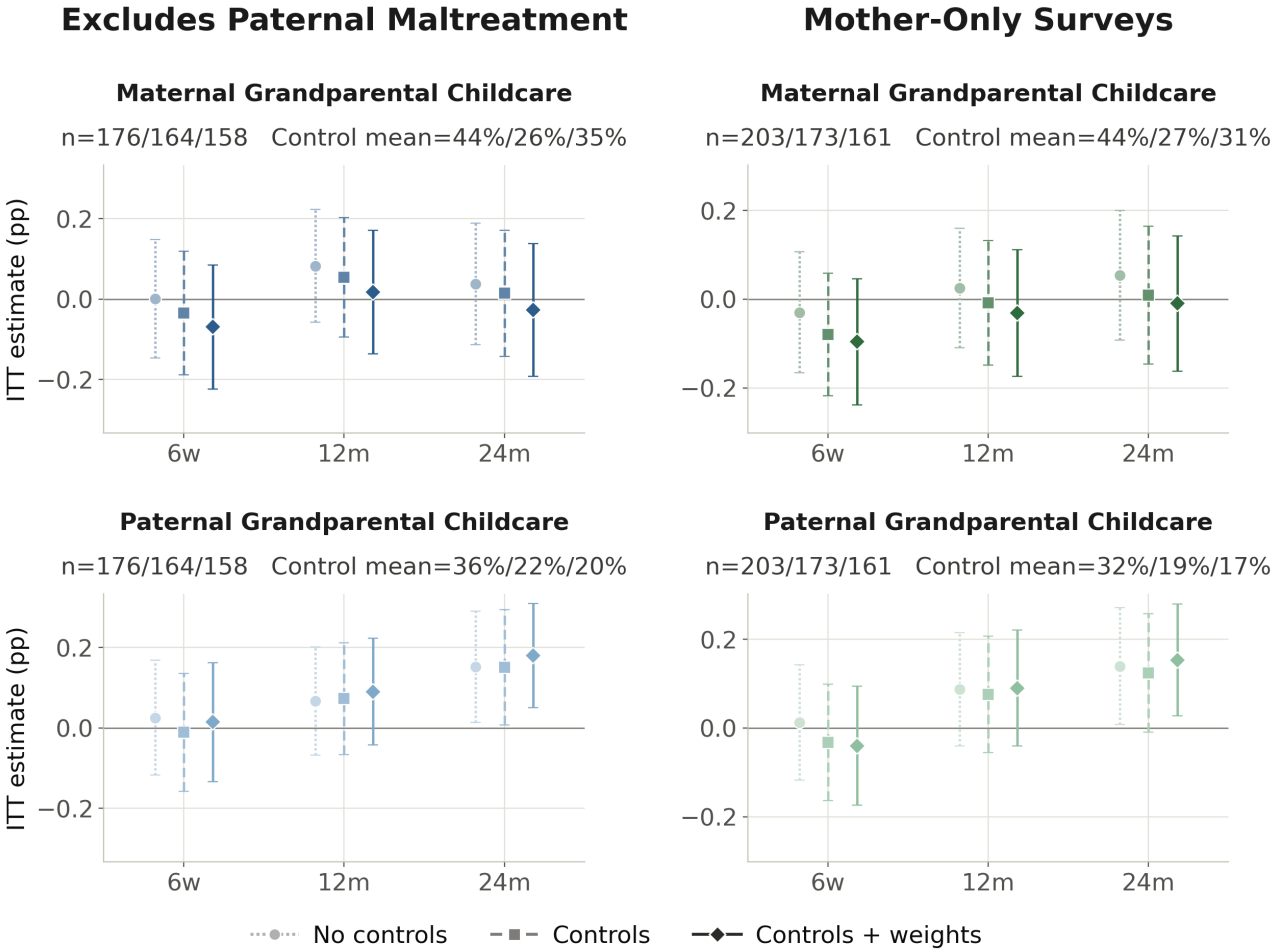
Notes: This figure shows the effect of Supportive Parenting on three types of contact: (1) contact with other parents, (2) contact with friends, (3) contact with neighbours. The figure shows ITT effects and their 95% confidence intervals from specifications without controls, with controls, and with controls and non-random attrition weights.

Table E.10: Robustness Table

	Controls 1			Controls 2			Controls 3		
	Control Mean	Treatment - Control	N	Control Mean	Treatment - Control	N	Control Mean	Treatment - Control	N
				Maternal Grandparental Childcare					
6 weeks	0.431	−0.077	214	0.438	−0.070	206	0.431	−0.057	214
	[0.498]	(0.067)		[0.499]	(0.076)		[0.498]	(0.068)	
12 months	0.272	−0.003	196	0.283	−0.012	189	0.272	−0.003	196
	[0.447]	(0.069)		[0.453]	(0.071)		[0.447]	(0.068)	
24 months	0.320	−0.005	189	0.319	0.021	182	0.320	0.000	189
	[0.469]	(0.070)		[0.469]	(0.071)		[0.469]	(0.069)	
				Paternal Grandparental Childcare					
6 weeks	0.321	−0.020	214	0.324	0.022	206	0.321	−0.020	214
	[0.469]	(0.067)		[0.470]	(0.071)		[0.469]	(0.066)	
12 months	0.204	0.088	196	0.212	0.089	189	0.204	0.079	196
	[0.405]	(0.063)		[0.411]	(0.068)		[0.405]	(0.064)	
24 months	0.175	0.160***	189	0.181	0.172***	182	0.175	0.158***	189
	[0.382]	(0.060)		[0.387]	(0.063)		[0.382]	(0.060)	
				Total Employment (Y1-Y3)					
Mother	24.495	−0.464	217	24.402	1.565	209	24.495	−0.255	217
	[14.442]	(1.323)		[14.389]	(1.600)		[14.442]	(1.312)	
Father	32.804	2.776**	209	32.804	1.459**	209	32.804	2.280**	209
	[7.632]	(1.096)		[7.632]	(0.730)		[7.632]	(0.883)	
Paternal grandmother	8.054	−2.392	187	8.054	−2.201	187	8.054	−2.551	187
	[14.579]	(2.244)		[14.579]	(2.173)		[14.579]	(2.150)	
Paternal grandfather	9.443	2.419	175	9.443	3.081	175	9.443	2.192	175
	[14.952]	(2.327)		[14.952]	(2.187)		[14.952]	(2.236)	

Notes: This table shows the estimated effects of SP on grandparental childcare and total employment for different sets of control variables and weights. For each group, the mean of the control group (Control Mean) and the estimated effect from a regression with controls and non-random attrition weights (Treatment - Control) are displayed. Controls 1 includes only maternal controls (age at birth, employment, benefits receipt, college), Controls 2 includes the regular set of controls without substitution for missing values, and Controls 3 includes the regular set of controls with rebalancing on the first moment only. Standard deviations are displayed in square brackets, robust standard errors are displayed in round parentheses, *p<0.10, **p<0.05, ***p<0.01.

Figure E.6: Robustness Figures



Notes: This figure shows the effect of Supportive Parenting on maternal and paternal grandparental childcare for two robustness checks: (1) excluding families where the father experienced maltreatment, (2) excluding families where the father filled out the survey. The figure shows ITT effects and their 95% confidence intervals from specifications without controls, with controls, and with controls and non-random attrition weights.

F Self-Reports

Table F.11: Balancing Table Attrition

	Complete at 6 weeks		Complete at 12 months		Complete at 24 months	
	Control Mean	Treatment - Control	Control Mean	Treatment - Control	Control Mean	Treatment - Control
Response rate	≥ 0.9	0.009	≥ 0.9	−0.051	0.874	−0.006
		(0.016)		(0.040)		(0.046)
			A. Mother			
Age at birth	32.552	−1.271**	32.665	−1.371**	32.814	−1.409**
	[4.299]	(0.615)	[4.072]	(0.612)	[4.147]	(0.622)
College degree	0.248	−0.076	0.272	−0.089	0.268	−0.072
		(0.056)		(0.060)		(0.061)
Employed	0.661	0.006	0.689	−0.023	0.680	0.015
		(0.065)		(0.067)		(0.068)
Benefits	0.220	−0.030	0.204	−0.010	0.196	−0.011
		(0.055)		(0.057)		(0.057)
Cohabitation	≥ 0.9	−0.031	≥ 0.9	−0.018	≥ 0.9	−0.025
		(0.037)		(0.038)		(0.038)
Observations	109	105	103	93	97	92
			B. Father			
Age at birth	35.399	−1.089	35.185	−0.953	35.831	−1.526*
	[5.846]	(0.762)	[5.615]	(0.782)	[5.782]	(0.804)
College degree	0.219	−0.041	0.212	−0.034	0.223	−0.019
		(0.056)		(0.058)		(0.061)
Employed	≥ 0.9	0.046	≥ 0.9	0.037	≥ 0.9	0.052
		(0.031)		(0.032)		(0.032)
Benefits	≤ 0.1	−0.027	≤ 0.1	−0.016	≤ 0.1	−0.018
		(0.031)		(0.033)		(0.034)
Observations	105	101	99	90	94	88
			C. Child			
Birth month	7.312	−0.160	7.301	−0.269	7.186	−0.012
	[3.884]	(0.523)	[4.017]	(0.548)	[4.053]	(0.565)
Firstborn	0.404	0.015	0.408	0.044	0.433	0.013
		(0.068)		(0.071)		(0.073)
Male	0.615	−0.043	0.602	−0.043	0.588	−0.012
		(0.067)		(0.071)		(0.072)
Observations	109	105	103	93	97	92

Notes: This table shows the response rate paternal grandparental childcare at 6 weeks, 12 months, and 24 months. This table also shows the difference between the control and treatment group mean for variables estimated in the month of birth for families eligible due to maternal maltreatment. The balancing table is shown for families with no missing data on paternal grandparental childcare at 6 weeks, at 12 months, and at 24 months. Standard deviations are displayed in square brackets, robust standard errors are displayed in round parentheses, *p<0.10, **p<0.05, ***p<0.01. Note that, in accordance with the privacy regulations of Statistics Netherlands, potentially disclosive statistics are not reported.

G Power Calculation

This appendix reports minimum detectable effects (MDEs) for the analysis sample, for rare outcomes, and for families eligible due to paternal childhood maltreatment. The MDE is the smallest true effect that the design detects with a probability of 80% at a two-sided significance level of 5%. For an estimator that is approximately normally distributed, the MDE is

$$\text{MDE} = (z_{0.975} + z_{0.80}) \times \text{SE}(\hat{\beta}_1) \approx 2.80 \times \text{SE}(\hat{\beta}_1),$$

where $\mathrm{SE}(\hat{\beta}_1)$ is the standard error of the estimated treatment effect.

I report two versions of the MDE, which differ only in how the standard error is obtained. The analytic MDE uses the standard error of a simple difference in means, calculated from the control group variance and the number of families in each group:

$$\mathrm{SE}_{\text{analytic}} = \sqrt{\sigma^2 \left(\frac{1}{n_c} + \frac{1}{n_t} \right)},$$

where $\sigma^2 = p_0(1 - p_0)$ for binary outcomes with control group mean p_0, and σ^2 is the control group variance for continuous outcomes. This is the MDE that could be calculated before estimating any regression. The ex-post MDE uses the robust standard error from the main specification with X_{ij} and non-random attrition weights. It therefore accounts for the precision gained from control variables and the precision lost due to weighting. For survey outcomes, n_c and n_t are based on the observed response rates of 90.3% at 12 months and 87.1% at 24 months.

Main analysis sample

Table G.12 reports MDEs for the analysis sample ($N = 217$). For grandparental childcare, the MDE is expressed as a change in the proportion of families receiving childcare. For employment, the MDE is expressed in months employed in the first three years after birth.

Table G.12: Minimum Detectable Effects in the Analysis Sample

Outcome	Wave	Control mean	N	MDE (analytic)	MDE (ex post)
A. Grandparental childcare (proportion; MDE as change in proportion)					
Paternal grandparents	12 months	0.204	196	0.161	0.190
Paternal grandparents	24 months	0.175	189	0.155	0.168
Maternal grandparents	12 months	0.272	196	0.178	0.191
Maternal grandparents	24 months	0.320	189	0.190	0.193
B. Months employed in first three years after birth (months; MDE in months)					
Mother		24.50	217	5.49	3.85
Father		32.80	209	2.96	2.05
Paternal grandmother		8.05	187	5.97	6.09
Paternal grandfather		9.44	175	6.33	6.13

Notes: MDEs for a two-sided significance level of 5% and power of 80%. In panel A, control means are proportions and an MDE of 0.161 corresponds to 16.1 percentage points. In panel B, control means and MDEs are in months, out of a maximum of 36. The analytic MDE equals 2.80 times the standard error of a simple difference in means, based on the control group variance and the number of families in each group. The ex-post MDE equals 2.80 times the robust standard error from the regression with X_{ij} and non-random attrition weights. For paternal grandparents, equal group sizes are assumed in the analytic MDE.

Rare outcomes

For outcomes with a low prevalence, the absolute MDE decreases, but the MDE relative to the control group mean increases sharply. Table G.13 reports analytic MDEs for hypothetical binary outcomes with a control group prevalence between 2.5% and 20%, using the 24-month sample sizes ($n_c = 97$, $n_t = 92$). Because the normal approximation underlying the analytic MDE is inaccurate when few events are observed, I also report MDEs based on 4,000 simulations per candidate effect size using Fisher's exact test. The last column expresses the Fisher exact MDE relative to the control group prevalence. With a prevalence of 5%, fewer than five control families are expected to report the outcome, and the programme would need to increase prevalence by around 13 percentage points, an increase of 260% relative to the control group prevalence, to be detected with 80% power. I therefore do not analyse rare outcomes such as financial transfers from grandparents or residential proximity.

Table G.13: Minimum Detectable Effects for Rare Binary Outcomes

Control prevalence	Expected events in control group	MDE (analytic)	MDE (Fisher exact)	Fisher exact MDE / control prevalence
	(1)	(2)	(3)	(4) = (3) / prevalence
0.025	2.4	0.064	0.11	440%
0.050	4.9	0.089	0.13	260%
0.100	9.7	0.122	0.16	160%
0.200	19.4	0.163	0.19	95%

Notes: MDEs for a two-sided significance level of 5% and power of 80%, with 97 control and 92 treatment families (the 24-month response rate). Control prevalence and MDEs in columns (2) and (3) are proportions. The analytic MDE equals 2.80 times the standard error of a difference in proportions. The Fisher exact MDE is the smallest increase, on a grid of one percentage point, for which at least 80% of 4,000 simulated trials reject the null hypothesis. Column (4) divides the Fisher exact MDE by the control group prevalence.

Families eligible due to paternal maltreatment

The sample of randomised families eligible due to paternal childhood maltreatment consists of 113 families. I assume equal allocation to treatment and control, the same response rates and grandparent linkage rates as in the analysis sample, and the same control group means and standard deviations. The ex-post MDE is approximated by multiplying the robust standard errors from the analysis sample by $\sqrt{N_{\text{analysis}}/N_{\text{paternal}}}$. Table G.14 reports the MDEs for this sample. For grandparental childcare, the analytic MDE ranges from 0.215 to 0.263 and the ex-post MDE from 0.233 to 0.268, corresponding to 21.5 to 26.8 percentage points. For employment, the analytic MDE is 7.6 months for mothers and 4.0 months for fathers, and the ex-post MDE is 5.3 months for mothers and 2.8 months for fathers. For paternal grandparents, the MDE ranges from 8.3 to 8.8 months across both versions. These MDEs are approximately 40% larger than in the analysis sample. I therefore do not analyse the effects of Supportive

Parenting in families eligible due to paternal maltreatment.

Table G.14: Minimum Detectable Effects in the Paternal Maltreatment Sample

Outcome	Wave	Control mean	N	MDE (analytic)	MDE (ex post)
A. Grandparental childcare (proportion; MDE as change in proportion)					
Paternal grandparents	12 months	0.204	102	0.223	0.263
Paternal grandparents	24 months	0.175	98	0.215	0.233
Maternal grandparents	12 months	0.272	102	0.247	0.264
Maternal grandparents	24 months	0.320	98	0.263	0.268
B. Months employed in first three years after birth (months; MDE in months)					
Mother		24.50	113	7.61	5.34
Father		32.80	113	4.02	2.78
Paternal grandmother		8.05	97	8.28	8.44
Paternal grandfather		9.44	91	8.78	8.49

Notes: MDEs for a two-sided significance level of 5% and power of 80%. In panel A, control means are proportions and MDEs are changes in proportions. In panel B, control means and MDEs are in months, out of a maximum of 36. Assumes 1:1 allocation of the 113 families, response rates and grandparent linkage rates equal to those in the analysis sample, and control group means and standard deviations equal to those in the analysis sample. The analytic MDE equals 2.80 times the standard error of a simple difference in means. The ex-post MDE multiplies the robust standard error from the regression with X_{ij} and non-random attrition weights in the analysis sample by $\sqrt{N_{\text{analysis}}/N_{\text{paternal}}}$.